

Calculus II

Homework

Area locked by curve

1. Let $x \in (0, 1)$. Express the following using x and $\sqrt{1 - x^2}$.

(a) $\sin(\arcsin(x))$.

(e) $\sin(2 \arccos(x))$.

(b) $\sin(2 \arcsin(x))$.

(f) $\sin(3 \arccos(x))$.

(c) $\sin(3 \arcsin(x))$.

(g) $\cos(2 \arcsin(x))$.

(d) $\sin(\arccos(x))$.

(h) $\cos(3 \arccos(x))$.

Solution. 1.b. Let $y = \arcsin x$. Then $\sin y = x$, and we can draw a right triangle with opposite side length x and hypotenuse length 1 to find the other trigonometric ratios of y .

Then $\cos y = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2}$. Now we use the double angle formula to find $\sin(2 \arcsin x)$.

$$\begin{aligned}\sin(2 \arcsin x) &= \sin(2y) \\ &= 2 \sin y \cos y \\ &= 2x\sqrt{1-x^2}.\end{aligned}$$

Solution. 1.c. Use the result of Problem 1.b. This also requires the addition formula for sine:

$$\sin(A + B) = \sin A \cos B + \sin B \cos A,$$

and the double angle formula for cosine:

$$\cos(2y) = \cos^2 y - \sin^2 y.$$

$$\begin{aligned}\sin(3 \arcsin x) &= \sin(3y) \\ &= \sin(2y + y) \\ &= \sin(2y) \cos y + \sin y \cos(2y) && \left| \begin{array}{l} \text{Use addition formula} \\ \text{Use double angle formulas} \end{array} \right. \\ &= (2 \sin y \cos y) \cos y + \sin y (\cos^2 y - \sin^2 y) \\ &= 2 \sin y \cos^2 y + \sin y \cos^2 y - \sin^3 y \\ &= 3 \sin y \cos^2 y - \sin^3 y \\ &= 3 \sin y (1 - \sin^2 y) - \sin^3 y \\ &= 3x(1 - x^2) - x^3 \\ &= 3x - 4x^3.\end{aligned}$$

The solution is complete. A careful look at the solution above reveals a strategy useful for problems similar to this one.

- Identify the inverse trigonometric expression- $\arcsin x, \arccos x, \arctan x, \dots$. In the present problem that was $y = \arcsin x$.
- The problem is therefore a trigonometric function of y .
- Using trig identities and algebra, rewrite the problem as a trigonometric expression involving only the trig function that transforms y to x . In the present problem we rewrote everything using $\sin y$.
- Use the fact that $\sin(\arcsin x) = x, \cos(\arccos x) = x, \dots$, etc. to simplify.

Solution. 1.f We use the same strategy outlined in the end of the solution of Problem 1.c. Set $y = \arccos x$ and so $\cos(y) = x$. Therefore:

$$\begin{aligned}\sin(3y) &= \sin(2y + y) \\ &= \sin(2y) \cos y + \sin y \cos(2y) \\ &= 2 \sin y \cos y \cos y + \sin y (2 \cos^2 y - 1) \\ &= 2 \sin y \cos^2 y + \sin y (2 \cos^2 y - 1) \\ &= \sin y (4 \cos^2 y - 1) && \left| \begin{array}{l} \text{use } \cos y = x \\ \sin y = \sqrt{1-x^2} \end{array} \right. \\ &= \sqrt{1-x^2} (4x^2 - 1) .\end{aligned}$$

2. Express as the following as an algebraic expression of x . In other words, “get rid” of the trigonometric and inverse trigonometric expressions.

(a) $\cos^2(\arctan x)$.

(b) $-\sin^2(\operatorname{arccot} x)$.

(c) $\frac{1}{\cos(\arcsin x)}$.

$$\frac{x^2+1}{1}$$

$$\frac{x^2+1}{1}$$

(d) $-\frac{1}{\sin(\arccos x)}$.

$$\frac{x^2-1}{1}$$

$$\frac{x^2-1}{1}$$

Solution. 2.b. We follow the strategy outlined in the end of the solution of Problem 1.c. We set $y = \operatorname{arccot} x$. Then we need to express $-\sin^2 y$ via $\cot y$. That is a matter of algebra:

$$\begin{aligned}
 -\sin^2(\operatorname{arccot} x) &= -\sin^2 y && \left| \begin{array}{l} \text{Set } y = \operatorname{arccot} x \\ \text{use } \sin^2 y + \cos^2 y = 1 \end{array} \right. \\
 &= -\frac{\sin^2 y}{\sin^2 y + \cos^2 y} \\
 &= -\frac{\sin^2 y}{1} \\
 &= -\frac{\sin^2 y + \cos^2 y}{\sin^2 y} \\
 &= -\frac{1}{1 + \cot^2 y} && \left| \begin{array}{l} \text{Substitute back } \cot y = x \end{array} \right. \\
 &= -\frac{1}{1 + x^2} .
 \end{aligned}$$

3. Rewrite as a rational function of t . This problem will be later used to derive the Euler substitutions (an important technique for integrating).

(a) $\cos(2 \arctan t)$.

(g) $\cos(2 \operatorname{arccot} t)$.

(b) $\sin(2 \arctan t)$.

(h) $\sin(2 \operatorname{arccot} t)$.

(c) $\tan(2 \arctan t)$.

(i) $\tan(2 \operatorname{arccot} t)$.

(d) $\cot(2 \arctan t)$.

(j) $\cot(2 \operatorname{arccot} t)$.

(e) $\csc(2 \arctan t)$.

(k) $\csc(2 \operatorname{arccot} t)$.

(f) $\sec(2 \arctan t)$.

(l) $\sec(2 \operatorname{arccot} t)$.

Solution. 3.a Set $z = \arctan t$, and so $\tan z = t$. Then

$$\begin{aligned}
 \cos(2 \arctan t) &= \cos(2z) \\
 &= \frac{\cos(2z)}{1} \\
 &= \frac{\cos^2 z - \sin^2 z}{\cos^2 z + \sin^2 z} && \left| \begin{array}{l} \text{use double angle formulas} \\ \text{and } 1 = \sin^2 z + \cos^2 z \end{array} \right. \\
 &= \frac{(\cos^2 z - \sin^2 z) \frac{1}{\cos^2 z}}{(\cos^2 z + \sin^2 z) \frac{1}{\cos^2 z}} \\
 &= \frac{1 - \tan^2 z}{1 + \tan^2 z} \\
 &= \frac{1 - t^2}{1 + t^2} .
 \end{aligned}$$

Solution. 3.d Set $z = \arctan t$, and so $\tan z = t$. Then

$$\begin{aligned}
 \cot(2 \arctan t) &= \cot(2z) \\
 &= \frac{\cos(2z)}{\sin(2z)} && \left| \begin{array}{l} \text{use double angle formulas} \end{array} \right. \\
 &= \frac{\cos^2 z - \sin^2 z}{2 \sin z \cos z} \\
 &= \frac{1 - \tan^2 z}{2 \tan z} \\
 &= \frac{1 - t^2}{2t} \\
 &= \frac{1}{2} \left(\frac{1}{t} - t \right) .
 \end{aligned}$$

$$(i) \int x^2 e^{2x} dx. \quad \text{ANSWER: } \frac{x^2}{2} \cos(3x) + \frac{6}{1} \sin(3x) + C$$

$$(j) \int x^3 e^x dx. \quad \text{ANSWER: } x^3 e^x - 3x^2 e^x + 6x e^x - 6e^x + C$$

$$\text{ANSWER: } \frac{x^2}{2} e^{\frac{x}{2}} + \frac{1}{x^2} e^{\frac{x}{2}} - \frac{2}{x} e^{\frac{x}{2}} + C$$

Solution. 6.a.

$$\int x \underbrace{\sin x dx}_{=d(-\cos x)} = - \int x d(\cos x) = -x \cos x + \int \cos x dx = -x \cos x + \sin x + C.$$

Solution. 6.c.

$$\begin{aligned} \int x^2 \underbrace{e^x dx}_{d(e^x)} &= \int x^2 de^x = x^2 e^x - \int e^x 2x dx = x^2 e^x - \int 2x de^x \\ &= x^2 e^x - 2x e^x + \int 2e^x dx = x^2 e^x - 2x e^x + 2e^x + C. \end{aligned}$$

Solution. 6.f.

$$\begin{aligned} \int x^2 e^{-2x} dx &= \int x^2 d\left(\frac{e^{-2x}}{-2}\right) && \left| \text{Integrate by parts} \right. \\ &= -\frac{x^2 e^{-2x}}{2} - \int \left(\frac{e^{-2x}}{-2}\right) d(x^2) \\ &= -\frac{x^2 e^{-2x}}{2} + \int x e^{-2x} dx \\ &= -\frac{x^2 e^{-2x}}{2} + \int x d\left(\frac{e^{-2x}}{-2}\right) && \left| \text{Integrate by parts} \right. \\ &= -\frac{x^2 e^{-2x}}{2} - \frac{x e^{-2x}}{2} + \frac{1}{2} \int e^{-2x} dx \\ &= -\frac{x^2 e^{-2x}}{2} - \frac{x e^{-2x}}{2} - \frac{e^{-2x}}{4} + C. \end{aligned}$$

7. Evaluate the indefinite integral. Illustrate the steps of your solutions.

$$(a) \int x^2 \cos(2x) dx. \quad \text{ANSWER: } \frac{x^2}{2} \sin(2x) + \frac{1}{2} \cos(2x) + C$$

$$(b) \int x^2 e^{ax} dx, \text{ where } a \text{ is a constant.} \quad \text{ANSWER: } \frac{x^2}{2} e^{ax} - \frac{x}{a} e^{ax} + \frac{1}{a^2} e^{ax} + C$$

$$(c) \int x^2 e^{-ax} dx, \text{ where } a \text{ is a constant.} \quad \text{ANSWER: } -\frac{x^2}{2} e^{-ax} - \frac{x}{a} e^{-ax} + \frac{1}{a^2} e^{-ax} + C$$

$$(d) \int x^2 \frac{(e^{ax} + e^{-ax})^2}{4} dx, \text{ where } a \text{ is a constant.} \quad \text{ANSWER: } \frac{x^3}{12} \ln(2x) + \frac{x^2}{6} + C$$

$$(e) \int \frac{1}{\cos^2 x} dx. \quad \text{(Hint: This problem does not require integration by parts. What is the derivative of } \tan x?) \quad \text{ANSWER: } \tan x + C$$

$$(f) \int (\tan^2 x) dx. \quad \text{(Hint: This problem does not require integration by parts. We can use } \tan^2 x = \frac{1}{\cos^2 x} - 1 \text{ and the previous problem.)} \quad \text{ANSWER: } x - \tan x + C$$

$$(g) \int x \tan^2 x dx. \quad \text{(Hint: } \tan^2 x dx = d(F(x)), \text{ where } F(x) \text{ is the answer from the preceding problem.)} \quad \text{ANSWER: } x \arcsin x + \sqrt{1-x^2} + C$$

$$(h) \int e^{-\sqrt{x}} dx. \quad \text{ANSWER: } -2\sqrt{x} e^{-\sqrt{x}} + 2e^{-\sqrt{x}} + C$$

$$(i) \int \cos^2 x dx. \quad \text{ANSWER: } \frac{x}{2} + \frac{\sin(2x)}{4} + C$$

$$(j) \int \frac{x}{1+x^2} dx \text{ (Hint: use substitution rule, don't use integration by parts)} \quad \text{ANSWER: } \frac{1}{2} \ln(1+x^2) + C$$

$$(k) \int (\arctan x) dx. \quad \text{ANSWER: } x \arctan x - \frac{1}{2} \ln(1+x^2) + C$$

$$(l) \int (\arcsin x) dx. \quad \text{ANSWER: } x \arcsin x + \sqrt{1-x^2} + C$$

$$(m) \int (\arcsin x)^2 dx. \quad \text{(Hint: Try substituting } x = \sin y.) \quad \text{ANSWER: } x(\arcsin x)^2 + 2\sqrt{1-x^2} \arcsin x - 2x + C$$

$$(n) \int \arctan\left(\frac{1}{x}\right) dx.$$

$$(o) \int \sin x e^x dx$$

$$(p) \int \cos x e^x dx$$

$$(q) \int \sin(\ln(x)) dx.$$

$$(r) \int \cos(\ln(x)) dx.$$

$$(s) \int \ln x dx$$

$$(t) \int x \ln x dx.$$

$$C + (x \sec^2 x - x \ln \sin x) \frac{e^x}{2} \quad \text{ANSWER}$$

$$(u) \int \frac{\ln x}{\sqrt{x}} dx.$$

$$C + (x \ln \ln x + x \cos x) \frac{e^x}{2} \quad \text{ANSWER}$$

$$(v) \int (\ln x)^2 dx.$$

$$C + ((x \ln \ln x) - (\sin \ln x) \cos \ln x) \frac{e^x}{2} \quad \text{ANSWER}$$

$$(w) \int (\ln x)^3 dx.$$

$$C + ((x \ln \ln x) + (\cos \ln x) \sin \ln x) \frac{e^x}{2} \quad \text{ANSWER}$$

$$(x) \int x^2 \cos^2 x dx. \quad (\text{This problem is related to Problem 7.d})$$

$$\text{as } \cos x = \frac{e^{ix} + e^{-ix}}{2}.$$

$$e^{ix} \frac{9}{4} + (x^2) \ln \frac{8}{4} - (x^2) \cos x \frac{5}{4} + (x^2) \ln \frac{5}{4} \quad \text{ANSWER}$$

Solution. 7.g.

$$\begin{aligned} \int x \tan^2 x dx &= \int x (\sec^2 x - 1) dx && \left| \begin{array}{l} \text{use } \sec^2 x - 1 = \tan^2 x \end{array} \right. \\ &= \int x (\sec^2 x - 1) dx \\ &= -\int x dx + \int x \sec^2 x dx && \left| \begin{array}{l} \text{use } d(\tan x) = \sec^2 x dx \end{array} \right. \\ &= -\frac{x^2}{2} + \int x d(\tan x) && \left| \begin{array}{l} \text{integrate by parts} \end{array} \right. \\ &= -\frac{x^2}{2} + x \tan x - \int \tan x dx \\ &= -\frac{x^2}{2} + x \tan x - \int \frac{\sin x}{\cos x} dx && \left| \begin{array}{l} \text{use } \sin x dx = -d(\cos x) \end{array} \right. \\ &= -\frac{x^2}{2} + x \tan x + \int \frac{d(\cos x)}{\cos x} && \left| \begin{array}{l} \text{Set } y = \cos x \end{array} \right. \\ &= -\frac{x^2}{2} + x \tan x + \int \frac{1}{y} dy \\ &= -\frac{x^2}{2} + x \tan x + \ln |y| + C && \left| \begin{array}{l} \text{Substitute back } y = \cos x \end{array} \right. \\ &= -\frac{x^2}{2} + x \tan x + \ln |\cos x| + C \quad . \end{aligned}$$

Solution. 7.h.

$$\begin{aligned} \int e^{-\sqrt{x}} dx &= \int 2ye^{-y} dy && \left| \begin{array}{l} \text{Subst.: } \sqrt{x} = y \\ \frac{1}{2\sqrt{x}} dx = dy \\ dx = 2y dy \end{array} \right. \\ &= \int 2y d(-e^{-y}) && \left| \begin{array}{l} \text{int. by parts} \end{array} \right. \\ &= -2ye^{-y} + 2 \int e^{-y} dy \\ &= -2ye^{-y} - 2e^{-y} + C \\ &= -2\sqrt{x}e^{-\sqrt{x}} - 2e^{-\sqrt{x}} + C \quad . \end{aligned}$$

Solution. 7.i. Later, we shall study general methods for solving trigonometric integrals that will cover this example. Let us however show one way to solve this integral by integration by parts.

$$\begin{aligned}
\int \cos^2 x dx &= x \cos^2 x - \int x d(\cos^2 x) \\
&= x \cos^2 x - \int x 2 \cos x (-\sin x) dx && \left| \sin(2x) = 2 \sin x \cos x \right. \\
&= x \cos^2 x + \int x \sin(2x) dx \\
&= x \cos^2 x + \int x d\left(\frac{-\cos(2x)}{2}\right) \\
&= x \cos^2 x + x \left(\frac{-\cos(2x)}{2}\right) - \int \left(\frac{-\cos(2x)}{2}\right) dx \\
&= \frac{x}{2} (2 \cos^2 x - \cos(2x)) + \frac{\sin(2x)}{4} + C && \left| \cos(2x) = \cos^2 x - \sin^2 x \right. \\
&= \frac{x}{2} (2 \cos^2 x - (\cos^2 x - \sin^2 x)) + \frac{\sin(2x)}{4} + C && \left| \cos^2 x + \sin^2 x = 1 \right. \\
&= \frac{x}{2} + \frac{\sin(2x)}{4} + C \quad .
\end{aligned}$$

Solution. 7.k

$$\begin{aligned}
\int \arctan x dx &= x \arctan x - \int x d(\arctan x) \\
&= x \arctan x - \int \frac{x}{x^2 + 1} dx \\
&= x \arctan x - \int \frac{\frac{1}{2} d(x^2)}{x^2 + 1} \\
&= x \arctan x - \int \frac{\frac{1}{2} d(x^2 + 1)}{x^2 + 1} \\
&= x \arctan x - \frac{1}{2} \ln(x^2 + 1) + C \quad .
\end{aligned}$$

Solution. 7.m.

$$\begin{aligned}
\int (\arcsin x)^2 dx &= \int (\arcsin(\sin y))^2 d(\sin y) && \left| \text{Set } x = \sin y \right. \\
&= \int y^2 \cos y dy = \int y^2 d(\sin y) && \left| \text{Integrate by parts} \right. \\
&= y^2 \sin y - \int 2y \sin y dy \\
&= y^2 \sin y + \int 2y d(\cos y) && \left| \text{Integrate by parts} \right. \\
&= y^2 \sin y + 2y \cos y - 2 \int \cos y dy \\
&= y^2 \sin y + 2y \cos y - 2 \sin y + C && \left| \text{Substitute } y = \arcsin x \right. \\
&= x(\arcsin x)^2 \\
&\quad + 2\sqrt{1 - x^2} \arcsin x - 2x + C \quad .
\end{aligned}$$

Solution. 7.o

$$\begin{aligned}
\int \sin x \underbrace{e^x dx}_{=de^x} &= \sin x e^x - \int e^x d(\sin x) = \sin x e^x - \int \cos x \underbrace{e^x dx}_{=de^x} \\
&= \sin x e^x - e^x \cos x + \int e^x d(\cos x) \\
&= e^x \sin x - e^x \cos x - \int e^x \sin x dx && \left| \text{add } \int e^x \sin x dx \text{ to both sides} \right. \\
2 \int \sin x e^x dx &= \sin x e^x - e^x \cos x \\
\int \sin x e^x dx &= \frac{1}{2} (\sin x e^x - e^x \cos x) \quad .
\end{aligned}$$

Solution. 7.q.

$$\begin{aligned}
\int \sin(\ln x) dx &= x \sin(\ln x) - \int x d(\sin(\ln x)) && \left| \begin{array}{l} \text{int. by parts} \end{array} \right. \\
&= x \sin(\ln x) - \int x (\cos(\ln x)) (\ln x)' dx \\
&= x \sin(\ln x) - \int \cos(\ln x) dx && \left| \begin{array}{l} \text{int. by parts} \end{array} \right. \\
&= x \sin(\ln x) - \left(x \cos(\ln x) - \int x d(\cos(\ln x)) \right) \\
&= x \sin(\ln x) - x \cos(\ln x) + \int x (-\sin(\ln x)) (\ln x)' dx \\
&= x \sin(\ln x) - x \cos(\ln x) - \int \sin(\ln x) dx && \left| \begin{array}{l} \text{add } \int \sin(\ln x) dx \\ \text{to both sides} \end{array} \right. \\
2 \int \sin(\ln x) dx &= x \sin(\ln x) - x \cos(\ln x) \\
\int \sin(\ln x) dx &= \frac{x}{2} (\sin(\ln x) - \cos(\ln x)) \quad .
\end{aligned}$$

Solution. 7.s

$$\int \ln x dx = x \ln x - \int x d(\ln x) = x \ln x - \int \frac{x}{x} dx = x \ln x - x + C \quad .$$

Solution. 7.u

$$\begin{aligned}
\int \frac{\ln x}{\sqrt{x}} dx &= \int (\ln x) 2d(\sqrt{x}) && \left| \begin{array}{l} \text{integrate by parts} \end{array} \right. \\
&= (\ln x) 2\sqrt{x} - \int 2\sqrt{x} d(\ln x) \\
&= 2\sqrt{x} \ln x - 2 \int \frac{\sqrt{x}}{x} dx \\
&= 2\sqrt{x} \ln x - 2 \int x^{-\frac{1}{2}} dx \\
&= 2\sqrt{x} \ln x - 4\sqrt{x} + C \\
&= 2\sqrt{x} (\ln x - 2) + C \quad .
\end{aligned}$$

8. Compute $\int x^n e^x dx$, where n is a non-negative integer.

Solution. 8

$$\begin{aligned}
\int x^n e^x dx &= \int x^n de^x \\
&= x^n e^x - \int e^x dx^n \\
&= x^n e^x - n \int x^{n-1} e^x dx \\
&= x^n e^x - n \left(\int x^{n-1} de^x \right) \\
&= x^n e^x - n \left(x^{n-1} e^x - \int (n-1) x^{n-2} e^x dx \right) \\
&= x^n e^x - n x^{n-1} e^x + n(n-1) \int x^{n-2} e^x dx \\
&= \dots (\text{continue above process}) \dots \\
&= x^n e^x - n x^{n-1} e^x + n(n-1) x^{n-2} e^x + \dots \\
&\quad + (-1)^k n(n-1)(n-2) \dots (n-k+1) x^{n-k} e^x \\
&\quad + \dots + (-1)^n n! e^x + C \\
&= C + \sum_{k=0}^n (-1)^k \frac{n!}{(n-k)!} x^{n-k} e^x \quad .
\end{aligned}$$

9. Integrate. Illustrate the steps of your solution.

$$(a) \int \frac{1}{x+1} dx$$

$$(b) \int \frac{x-1}{x+1} dx$$

$$(c) \int \frac{1}{(x+1)^2} dx$$

$$(d) \int \frac{x}{(x+1)^2} dx$$

$$(e) \int \frac{1}{(2x+3)^2} dx$$

$$(f) \int \frac{x}{2x^2+3} dx$$

$$(g) \int \frac{1}{2x^2+3} dx$$

$$(h) \int \frac{x}{2x^2+x+1} dx$$

$$(i) \int \frac{x}{2x^2+x+3} dx$$

$$(j) \int \frac{x}{x^2-x+3} dx$$

$$(k) \int \frac{1}{(x^2+1)^2} dx$$

$$(l) \int \frac{1}{(x^2+x+1)^2} dx$$

$$(m) \int \frac{1}{(x^2+1)^3} dx$$

Solution. 9.h.

$$\begin{aligned} \int \frac{x}{2x^2+x+1} dx &= \int \frac{x}{2\left(x^2+2x\frac{1}{4}+\frac{1}{2}\right)} dx \\ &= \int \frac{x}{2\left(x^2+2x\frac{1}{4}+\frac{1}{16}-\frac{1}{16}+\frac{1}{2}\right)} dx && \left| \begin{array}{l} \text{complete square} \\ \text{in denominator} \end{array} \right. \\ &= \frac{1}{2} \int \frac{x}{\left(x+\frac{1}{4}\right)^2+\frac{7}{16}} dx \\ &= \frac{1}{2} \int \frac{x+\frac{1}{4}-\frac{1}{4}}{\left(x+\frac{1}{4}\right)^2+\frac{7}{16}} d\left(x+\frac{1}{4}\right) && \left| \begin{array}{l} \text{Set } u = x + \frac{1}{4} \end{array} \right. \\ &= \frac{1}{2} \int \frac{u-\frac{1}{4}}{u^2+\frac{7}{16}} du \\ &= \frac{1}{2} \left(\int \frac{u}{u^2+\frac{7}{16}} du - \frac{1}{4} \int \frac{1}{u^2+\frac{7}{16}} du \right) \\ &= \frac{1}{2} \left(\frac{1}{2} \ln \left(u^2 + \frac{7}{16} \right) - \frac{1}{4\sqrt{\frac{7}{16}}} \arctan \left(\frac{u}{\sqrt{\frac{7}{16}}} \right) \right) + K \\ &= \frac{1}{4} \ln \left(x^2 + \frac{1}{2}x + \frac{1}{2} \right) - \frac{\sqrt{7}}{14} \arctan \left(\frac{4x+1}{\sqrt{7}} \right) + K \end{aligned}$$

Solution. 9.i

$$\begin{aligned} \int \frac{1}{(x^2+x+1)^2} dx &= \int \frac{1}{\left(\left(x^2+2x\frac{1}{2}+\frac{1}{4}\right)-\frac{1}{4}+1\right)^2} dx && \left| \begin{array}{l} \text{complete the square} \end{array} \right. \\ &= \int \frac{1}{\left(\left(x+\frac{1}{2}\right)^2+\frac{3}{4}\right)^2} d\left(x+\frac{1}{2}\right) && \left| \begin{array}{l} \text{Set } w = x + \frac{1}{2} \end{array} \right. \\ &= \int \frac{1}{\left(w^2+\frac{3}{4}\right)^2} dw \\ &= \int \frac{1}{\left(\frac{3}{4}\left(\left(\frac{2w}{\sqrt{3}}\right)^2+1\right)\right)^2} \frac{\sqrt{3}}{2} d\left(\frac{2w}{\sqrt{3}}\right) && \left| \begin{array}{l} \text{Set } z = \frac{2w}{\sqrt{3}} \end{array} \right. \\ &= \frac{\frac{\sqrt{3}}{2}}{\left(\frac{3}{4}\right)^2} \int \frac{1}{(z^2+1)^2} dz \\ &= \frac{8\sqrt{3}}{9} \int \frac{1}{(z^2+1)^2} dz \end{aligned}$$

The integral $\int \frac{1}{(z^2+1)^2} dz$ was already studied; it was also given as an exercise in Problem 9.k. We leave the rest of the problem to the reader.

10. Let a, b, c, A, B be real numbers. Suppose in addition $a \neq 0$ and $b^2 - 4ac < 0$. Integrate

$$\int \frac{Ax + B}{ax^2 + bx + c} dx \quad .$$

The purpose of this exercise is to produce a formula in form ready for implementation in a computer algebra system.

Solution. 10.

$$\begin{aligned} \int \frac{Ax + B}{ax^2 + bx + c} dx &= \int \frac{Ax + B}{a \left(x^2 + 2x \frac{b}{2a} + \frac{c}{a} \right)} dx \\ &= \int \frac{Ax + B}{a \left(x^2 + 2x \frac{b}{2a} + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} + \frac{c}{a} \right)} dx && \left| \begin{array}{l} \text{complete square} \\ \text{in denominator} \end{array} \right. \\ &= \frac{1}{a} \int \frac{Ax + B}{\left(x + \frac{b}{2a} \right)^2 + \frac{4ac - b^2}{4a^2}} dx && \left| \begin{array}{l} \text{Set } D = \frac{4ac - b^2}{4a^2} \end{array} \right. \\ &= \frac{1}{a} \int \frac{A \left(x + \frac{b}{2a} - \frac{b}{2a} \right) + B}{\left(x + \frac{b}{2a} \right)^2 + D} d \left(x + \frac{b}{2a} \right) && \left| \begin{array}{l} \text{Set } u = x + \frac{b}{2a} \end{array} \right. \\ &= \frac{1}{a} \int \frac{Au + B - \frac{Ab}{2a}}{u^2 + D} du && \left| \begin{array}{l} \text{Set } C = B - \frac{Ab}{2a} \end{array} \right. \\ &= \frac{1}{a} \left(A \int \frac{u}{u^2 + D} du + C \int \frac{1}{u^2 + D} du \right) \\ &= \frac{1}{a} \left(\frac{A}{2} \ln(u^2 + D) + \frac{C}{\sqrt{D}} \arctan \left(\frac{u}{\sqrt{D}} \right) \right) + K \\ &= \frac{1}{a} \left(\frac{A}{2} \ln \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right) \right. \\ &\quad \left. + \frac{C}{\sqrt{D}} \arctan \left(\frac{x + \frac{b}{2a}}{\sqrt{D}} \right) \right) + K. \end{aligned}$$

The solution is complete. Question to the student: where do we use $b^2 - 4ac < 0$?

11. Let a, b, c, A, B be real numbers and let $n > 1$ be an integer. Suppose in addition $a \neq 0$ and $b^2 - 4ac < 0$. Let

$$J(n) = \int \frac{1}{\left(x^2 + \frac{b}{a}x + \frac{c}{a} \right)^n} dx \quad .$$

- (a) Express the integral

$$\int \frac{Ax + B}{(ax^2 + bx + c)^n} dx$$

via $J(n)$.

- (b) Express $J(n)$ recursively via $J(n-1)$

The purpose of this exercise is to produce a formula in form ready for implementation in a computer algebra system.

Solution. 11.a.

$$\begin{aligned}
\int \frac{Ax+B}{(ax^2+bx+c)^n} dx &= \int \frac{Ax+B}{a^n \left(x^2 + 2x \frac{b}{2a} + \frac{c}{a}\right)^n} dx \\
&= \int \frac{Ax+B}{a^n \left(x^2 + 2x \frac{b}{2a} + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} + \frac{c}{a}\right)^n} dx && \text{complete square in denominator} \\
&= \frac{1}{a^n} \int \frac{Ax+B}{\left(\left(x + \frac{b}{2a}\right)^2 + \frac{4ac-b^2}{4a^2}\right)^n} dx && \text{Set } D = \frac{4ac-b^2}{4a^2} \\
&= \frac{1}{a^n} \int \frac{A\left(x + \frac{b}{2a} - \frac{b}{2a}\right) + B}{\left(\left(x + \frac{b}{2a}\right)^2 + D\right)^n} d\left(x + \frac{b}{2a}\right) && \text{Set } u = x + \frac{b}{2a} \\
&= \frac{1}{a^n} \int \frac{Au + B - \frac{Ab}{2a}}{(u^2 + D)^n} du && \text{Set } C = B - \frac{Ab}{2a} \\
&= \frac{1}{a^n} \left(A \int \frac{u}{(u^2 + D)^n} du + C \int \frac{1}{(u^2 + D)^n} du \right) \\
&= \frac{1}{a^n} \left(\frac{A}{2(1-n)} (u^2 + D)^{1-n} + C J(n) \right) \\
&= \frac{1}{a^n} \left(\frac{A}{2(1-n)} \left(x^2 + \frac{b}{a}x + \frac{c}{a}\right)^{1-n} + C J(n) \right)
\end{aligned}$$

Solution. 11.b. We use all notation and computations from the previous part of the problem. According to theory, in order to solve that integral, we are supposed to integrate by parts the simpler integral

$$\begin{aligned}
J(n-1) &= \int \frac{1}{\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right)^{n-1}} dx = \int \frac{1}{(u^2 + D)^{n-1}} du && \text{int. by parts} \\
&= \frac{u}{(u^2 + D)^{n-1}} - \int u d\left(\frac{1}{(u^2 + D)^{n-1}}\right) \\
&= \frac{u}{(u^2 + D)^{n-1}} + 2(n-1) \int \frac{u^2}{(u^2 + D)^n} du \\
&= \frac{u}{(u^2 + D)^{n-1}} + 2(n-1) \int \frac{u^2 + D - D}{(u^2 + D)^n} du \\
&= \frac{u}{(u^2 + D)^{n-1}} + 2(n-1) J(n-1) - 2D(n-1) \int \frac{1}{(u^2 + D)^n} du \\
&= \frac{u}{(u^2 + D)^{n-1}} + 2(n-1) J(n-1) - 2D(n-1) J(n)
\end{aligned}$$

In the above equality, we rearrange

terms to get that

$$\begin{aligned}
2D(n-1)J(n) &= \frac{u}{(u^2 + D)^{n-1}} + (2n-3)J(n-1) \\
J(n) &= \frac{1}{D} \left(\frac{u}{2(n-1)(u^2 + D)^{n-1}} + \frac{2n-3}{2n-2} J(n-1) \right) \\
&= \frac{1}{D} \left(\frac{x + \frac{b}{2a}}{(2n-2)\left(x^2 + \frac{b}{a}x + \frac{c}{a}\right)^{n-1}} + \frac{2n-3}{2n-2} J(n-1) \right)
\end{aligned}$$

12. Integrate. Some of the examples require partial fraction decomposition and some do not. Illustrate the steps of your solution.

(a) $\int \frac{1}{4x^2 + 4x + 1} dx$

ANSWER: $-\frac{2}{1} \ln|1+x| + \frac{2}{1} \ln|1+x|$

(e) $\int \frac{x+1}{2x^2+x} dx$

ANSWER: $-\frac{2}{1} \ln|2x+1| + \frac{2}{1} \ln|x|$

(b) $\int \frac{1}{1-x^2} dx$

ANSWER: $-\frac{2}{1} \ln|1-x| + \frac{2}{1} \ln|1+x|$

(f) $\int \frac{x}{4x^2+x+5} dx$

ANSWER: $\frac{8}{1} \ln|x^2 + \frac{1}{4}x + \frac{5}{4}| - \frac{9}{1} \sqrt{\frac{64}{1} \arctan\left(\frac{64x+1}{8}\right)}$

(c) $\int \frac{1}{5-x^2} dx$

ANSWER: $-\frac{0.1}{2} \ln|2-x| + \frac{0.1}{2} \ln|2+x|$

(g) $\int \frac{x}{4x^2+x-5} dx$

ANSWER: $\frac{9}{2} \ln|x| + \frac{6}{1} \ln|1-x| + \frac{3}{2} \ln|2+x|$

(d) $\int \frac{x}{4x^2+x+\frac{1}{16}} dx$

ANSWER: $\frac{1}{1} \ln|8x+1| - \frac{1}{1} \ln|1+x|$

(h) $\int \frac{x}{3x^2+x-2} dx$

ANSWER: $\frac{9}{2} \ln|x| - \frac{3}{2} \ln|1+x| + \frac{5}{1} \ln|1-x|$

$$\begin{aligned}
\text{(i)} \quad \int \frac{x}{3x^2 + x + 2} dx &= \frac{1}{6} \ln \left| \frac{2x+1}{x+1} \right| + \frac{1}{6} \ln \left| \frac{x-2}{x+1} \right| + C \\
\text{(j)} \quad \int \frac{x}{2x^2 + x + 1} dx &= \frac{1}{2} \ln \left| \frac{x+1}{x-1} \right| + \frac{1}{2} \ln \left| \frac{x+1}{x+2} \right| + C \\
\text{(k)} \quad \int \frac{x}{2x^2 + x - 1} dx &= \frac{1}{3} \ln \left| \frac{x+1}{x-1} \right| + \frac{1}{6} \ln \left| \frac{x-1}{x+2} \right| + C \\
\text{(l)} \quad \int \frac{1}{x^2 + x + 1} dx &= \frac{2}{\sqrt{3}} \arctan \left(\frac{2x+1}{\sqrt{3}} \right) + C \\
\text{(m)} \quad \int \frac{1}{2x^2 + 5x + 1} dx &= \frac{1}{\sqrt{33}} \ln \left| \frac{2x+1}{x+2} \right| - \frac{1}{\sqrt{33}} \ln \left| \frac{2x+1}{x-1} \right| + C
\end{aligned}$$

Solution. 12.k The quadratic in the denominator has real roots and therefore can be factored using real numbers. We therefore use partial fractions.

$$\begin{aligned}
\int \frac{x}{2x^2 + x - 1} dx &= \int \frac{\frac{1}{2}x}{(x+1)(x-\frac{1}{2})} dx \quad \left| \text{partial fractions, see below} \right. \\
&= \int \frac{\frac{1}{3}}{(x+1)} dx + \int \frac{\frac{1}{6}}{(x-\frac{1}{2})} dx \\
&= \frac{1}{3} \ln |x+1| + \frac{1}{6} \ln \left| x - \frac{1}{2} \right| + C.
\end{aligned}$$

Except for showing how the partial fraction decomposition was obtained, our solution is complete. We proceed to compute the partial fraction decomposition used above.

We aim to decompose into partial fractions the following function (the denominator has been factored).

$$\frac{x}{2x^2 + x - 1} = \frac{x}{(x+1)(2x-1)} = \frac{A_1}{x+1} + \frac{A_2}{2x-1}.$$

After clearing denominators, we get the following equality.

$$x = A_1(2x-1) + A_2(x+1). \quad (2)$$

Next, we need to find values for A_1 and A_2 such that the equality above becomes an identity. We show two variants to do that: the method of substitutions and the method of coefficient comparison.

Variant I. This variant relies on the fact that if substitute an arbitrary value for x in (2) we get a relationship that must be satisfied by the coefficients A_1 and A_2 . We immediately see that setting $x = \frac{1}{2}$ (notice $x = \frac{1}{2}$ is a root of the denominator) will annihilate the term $A_1(2x-1)$ and we can immediately solve for A_2 . Similarly, setting $x = -1$ ($x = -1$ is the other root of the denominator) annihilates the term $A_2(x+1)$ and we can immediately solve for A_1 .

- Set $x = \frac{1}{2}$. The equation (2) becomes

$$\begin{aligned}
\frac{1}{2} &= A_1 \cdot 0 + A_2 \left(\frac{1}{2} + 1 \right) \\
\frac{1}{2} &= \frac{3}{2} A_2 \\
A_2 &= \frac{1}{3}.
\end{aligned}$$

- Set $x = -1$. The equation (2) becomes

$$\begin{aligned}
-1 &= A_1(2 \cdot (-1) - 1) + A_2 \cdot 0 \\
-1 &= -3A_1 \\
A_1 &= \frac{1}{3}.
\end{aligned}$$

Therefore we have the partial fraction decomposition

$$\begin{aligned}
\frac{x}{2x^2 + x - 1} &= \frac{A_1}{x+1} + \frac{A_2}{2x-1} \\
&= \frac{\frac{1}{3}}{x+1} + \frac{\frac{1}{3}}{2x-1} \\
&= \frac{\frac{1}{3}}{x+1} + \frac{\frac{1}{6}}{x-\frac{1}{2}}.
\end{aligned}$$

Variant II. We show the most straightforward technique for finding a partial fraction decomposition - the method of coefficient comparison. Although this technique is completely doable in practice by hand, it is often the most laborious for a human. We

note that techniques such as the one given in the preceding solution Variant are faster on many (but not all) problems. The present technique is also arguably the easiest to implement on a computer. The computations below were indeed carried out by a computer program written for the purpose.

After rearranging we get that the following polynomial must vanish. Here, by “vanish” we mean that the coefficients of the powers of x must be equal to zero.

$$(A_2 + 2A_1 - 1)x + (A_2 - A_1) = 0.$$

In other words, we need to solve the following system.

$$\begin{aligned} 2A_1 + A_2 &= 1 \\ -A_1 + A_2 &= 0 \end{aligned}$$

System status	Action
$2A_1 + A_2 = 1$ $-A_1 + A_2 = 0$	Sel. pivot column 2. Eliminate non-pivot entries.
$A_1 + \frac{A_2}{2} = \frac{1}{2}$ $\frac{3}{2}A_2 = \frac{1}{2}$	Sel. pivot column 3. Eliminate non-pivot entries.
$A_1 = \frac{1}{3}$ $A_2 = \frac{1}{3}$	Final result.

Therefore, the final partial fraction decomposition is:

$$\frac{x}{x^2 + \frac{x}{2} - \frac{1}{2}} = \frac{\frac{1}{3}}{(x+1)} + \frac{\frac{1}{3}}{(2x-1)}.$$

13. Evaluate the indefinite integral. Illustrate all steps of your solution.

(a) $\int \frac{x^3 + 4}{x^2 + 4} dx$

$$\frac{x}{2} + (4 + \frac{x}{2}) \ln |x - \frac{x}{2}| - \frac{1}{2} \ln |x^2 + 4| + C$$

(b) $\int \frac{4x^2}{2x^2 - 1} dx$

$$2x + \frac{1}{2} \ln |2x^2 - 1| + C$$

(c) $\int \frac{x^3}{x^2 + 2x - 3} dx$

$$\frac{x^2}{2} + \frac{1}{2} \ln |x^2 + 2x - 3| + C$$

(d) $\int \frac{x^3}{x^2 + 3x - 4} dx$

$$\frac{x^2}{2} + \frac{1}{2} \ln |x^2 + 3x - 4| + C$$

(e) $\int \frac{x^3}{2x^2 + 3x - 5} dx$

$$\frac{x^2}{2} + \frac{1}{2} \ln |2x^2 + 3x - 5| + C$$

(f) $\int \frac{x^2 + 1}{(x - 3)(x - 2)^2} dx$

$$\frac{1}{x - 3} + \frac{1}{x - 2} + \frac{1}{(x - 2)^2} + C$$

(g) $\int \frac{x^4}{(x + 1)^2(x + 2)} dx$

$$\frac{x^2}{2} + \frac{1}{2} \ln |x + 1| + \frac{1}{2} \ln |x + 2| + C$$

(h) $\int \frac{15x^2 - 4x - 81}{(x - 3)(x + 4)(x - 1)} dx$

$$\frac{1}{x - 3} + \frac{1}{x + 4} + \frac{1}{x - 1} + C$$

(i) $\int \frac{x^4 + 10x^3 + 18x^2 + 2x - 13}{x^4 + 4x^3 + 3x^2 - 4x - 4} dx$

Check first that $(x - 1)(x + 2)^2(x + 1) = x^4 + 4x^3 + 3x^2 - 4x - 4$.

(j) $\int \frac{x^4}{(x^2 + 2)(x + 2)} dx$

$$\frac{x^2}{2} + \frac{1}{2} \ln |x^2 + 2| + \frac{1}{2} \ln |x + 2| + C$$

(k) $\int \frac{x^5}{x^3 - 1} dx$

$$\frac{x^2}{2} + \frac{1}{2} \ln |x^3 - 1| + C$$

(l) $\int \frac{x^4}{(x^2 + 2)(x + 1)^2} dx$

$$\frac{x^2}{2} + \frac{1}{2} \ln |x^2 + 2| + \frac{1}{2} \ln |x + 1| + C$$

(m) $\int \frac{3x^2 + 2x - 1}{(x - 1)(x^2 + 1)} dx$

$$\frac{1}{x - 1} + \frac{1}{x^2 + 1} + C$$

(n) $\int \frac{x^2 - 1}{x(x^2 + 1)^2} dx$

$$\frac{1}{x} - \frac{1}{x^2 + 1} + C$$

Solution. 13.1 To integrate a rational function, we need to decompose it into partial fractions.

Since the numerator of the function is of degree greater than or equal to the denominator, we start the partial fraction decomposition by polynomial division.

	Remainder $-2x^3 \quad -3x^2 \quad -4x \quad -2$
Divisor(s) $x^4 + 2x^3 + 3x^2 + 4x + 2$	Quotient(s) 1
	Dividend x^4 $x^4 \quad +2x^3 \quad +3x^2 \quad +4x \quad +2$ $-2x^3 \quad -3x^2 \quad -4x \quad -2$

Our next step is to factor the denominator:

$$x^4 + 2x^3 + 3x^2 + 4x + 2 = (x + 1)^2 (x^2 + 2).$$

Next, we combine the two steps:

$$\begin{aligned} \frac{x^4}{x^4 + 2x^3 + 3x^2 + 4x + 2} &= 1 + \frac{-2x^3 - 3x^2 - 4x - 2}{x^4 + 2x^3 + 3x^2 + 4x + 2} \\ &= \frac{-2x^3 - 3x^2 - 4x - 2}{x^4 + 2x^3 + 3x^2 + 4x + 2} \\ &= \frac{(x + 1)^2 (x^2 + 2)}{(x + 1)^2 (x^2 + 2)} \\ &= \frac{A_1}{(x + 1)} + \frac{A_2}{(x + 1)^2} + \frac{A_3 + A_4 x}{(x^2 + 2)}. \end{aligned}$$

We seek to find A_i 's that turn the above expression into an identity. Just as in the solution of Problem 12.k, we will use the method of coefficient comparison. We note that the solutions of Problems 13.m and 12.k provide a shortcut method.

After clearing denominators, we get the following equality.

$$\begin{aligned} -2x^3 - 3x^2 - 4x - 2 &= A_1(x + 1)(x^2 + 2) + A_2(x^2 + 2) \\ &\quad + (A_3 + A_4 x)(x + 1)^2 \\ 0 &= (A_4 + A_1 + 2)x^3 \\ &\quad + (2A_4 + A_3 + A_2 + A_1 + 3)x^2 \\ &\quad + (A_4 + 2A_3 + 2A_1 + 4)x \\ &\quad + (A_3 + 2A_2 + 2A_1 + 2). \end{aligned}$$

In order to turn the above into an identity we need to select A_i 's such that the coefficients of all powers of x become zero. In other words, we need to solve the following system.

$$\begin{aligned} A_1 &\quad \quad \quad + A_4 &= -2 \\ A_1 &\quad + A_2 &\quad + A_3 &\quad + 2A_4 &= -3 \\ 2A_1 &\quad \quad \quad + 2A_3 &\quad + A_4 &= -4 \\ 2A_1 &\quad + 2A_2 &\quad + A_3 & &= -2 \end{aligned}$$

This is a system of linear equations. There exists a standard method for solving such systems called Gaussian Elimination (this method is also known as the row-echelon form reduction method). This method is very well suited for computer implementation. We illustrate it on this particular example; for a description of the method in full generality we direct the reader to a standard course in Linear algebra.

System status	Action
$A_1 \quad \quad \quad + A_4 = -2$	
$A_1 \quad + A_2 \quad + A_3 \quad + 2A_4 = -3$	
$2A_1 \quad \quad \quad + 2A_3 \quad + A_4 = -4$	Sel. pivot column 2. Eliminate non-pivot entries.
$2A_1 \quad + 2A_2 \quad + A_3 \quad \quad \quad = -2$	
$A_1 \quad \quad \quad + A_4 = -2$	
$A_2 \quad + A_3 \quad + A_4 = -1$	
$2A_3 \quad - A_4 = 0$	Sel. pivot column 3. Eliminate non-pivot entries.
$2A_2 \quad + A_3 \quad - 2A_4 = 2$	

A_1		$+A_4$	$= -2$	Sel. pivot column 4. Eliminate non-pivot entries.	
	A_2	$+A_3$	$+A_4$		$= -1$
		$2A_3$	$-A_4$		$= 0$
		$-A_3$	$-4A_4$		$= 4$
A_1		$+A_4$	$= -2$	Sel. pivot column 5. Eliminate non-pivot entries.	
	A_2	$+\frac{3}{2}A_4$	$= -1$		
	A_3	$-\frac{A_4}{2}$	$= 0$		
		$-\frac{9}{2}A_4$	$= 4$		
A_1			$= -\frac{10}{9}$	Final result.	
	A_2		$= \frac{1}{3}$		
	A_3		$= -\frac{4}{9}$		
	A_4		$= -\frac{8}{9}$		

Therefore, the final partial fraction decomposition is the following.

$$\begin{aligned}
 \frac{x^4}{x^4 + 2x^3 + 3x^2 + 4x + 2} &= 1 + \frac{-2x^3 - 3x^2 - 4x - 2}{x^4 + 2x^3 + 3x^2 + 4x + 2} \\
 &= 1 + \frac{-\frac{10}{9}}{(x+1)} + \frac{\frac{1}{3}}{(x+1)^2} + \frac{-\frac{8}{9}x - \frac{4}{9}}{(x^2+2)}
 \end{aligned}$$

Therefore we can integrate as follows.

$$\begin{aligned}
 \int \frac{x^4}{(x^2+2)(x+1)^2} dx &= \int \left(1 + \frac{-\frac{10}{9}}{(x+1)} + \frac{\frac{1}{3}}{(x+1)^2} + \frac{-\frac{8}{9}x - \frac{4}{9}}{(x^2+2)} \right) dx \\
 &= \int dx - \frac{10}{9} \int \frac{1}{(x+1)} dx + \frac{1}{3} \int \frac{1}{(x+1)^2} dx \\
 &\quad - \frac{8}{9} \int \frac{x}{x^2+2} dx - \frac{4}{9} \int \frac{1}{x^2+2} dx \\
 &= x - \frac{1}{3}(x+1)^{-1} - \frac{10}{9} \log(x+1) \\
 &\quad - \frac{4}{9} \log(x^2+2) - \frac{2}{9} \sqrt{2} \arctan\left(\frac{\sqrt{2}}{2}x\right) + C
 \end{aligned}$$

Solution. 13.k This problem can be solved directly with a substitution shortcut, or by the standard method.

Variant I (standard method).

$ \begin{aligned} \int \frac{x^5}{x^3-1} dx &= \int \left(x^2 + \frac{x^2}{x^3-1} \right) dx \\ &= \frac{x^3}{3} + \int \frac{x^2}{(x-1)(x^2+x+1)} dx \\ &= \frac{x^3}{3} + \int \left(\frac{\frac{1}{3}}{x-1} + \frac{\frac{2}{3}x + \frac{1}{3}}{x^2+x+1} \right) dx \\ &= \frac{x^3}{3} + \frac{1}{3} \ln x-1 + \frac{2}{3} \int \frac{x + \frac{1}{2}}{(x + \frac{1}{2})^2 + \frac{3}{4}} dx \\ &= \frac{x^3}{3} + \frac{1}{3} \ln x-1 + \frac{1}{3} \int \frac{du}{u} \\ &= \frac{x^3}{3} + \frac{1}{3} \ln x-1 + \frac{1}{3} \ln u + C \\ &= \frac{x^3}{3} + \frac{1}{3} \ln x-1 + \frac{1}{3} \ln x^2+x+1 + C \end{aligned} $	Polyn. long div. part. frac. complete square Set $\begin{aligned} u &= (x + \frac{1}{2})^2 + \frac{3}{4} \\ \frac{1}{2} du &= (x + \frac{1}{2}) dx \end{aligned}$
--	--

Variant II (shortcut method).

$$\begin{aligned}
 \int \frac{x^5}{x^3-1} dx &= \int \frac{x^5 - x^2 + x^2}{x^3-1} dx \\
 &= \int \frac{x^2(x^3-1) + x^2}{x^3-1} dx \\
 &= \int x^2 dx + \int \frac{x^2}{x^3-1} dx \\
 &= \frac{x^3}{3} + \int \frac{d\left(\frac{x^3}{3}\right)}{x^3-1} \\
 &= \frac{x^3}{3} + \frac{1}{3} \int \frac{d(x^3-1)}{x^3-1} \quad \left| \text{Set } u = x^3 - 1 \right. \\
 &= \frac{x^3}{3} + \frac{1}{3} \int \frac{du}{u} \\
 &= \frac{x^3}{3} + \frac{1}{3} \ln |u| + C \\
 &= \frac{x^3}{3} + \frac{1}{3} \ln |x^3 - 1| + C .
 \end{aligned}$$

The answers obtained in the two solution variants are of course equal since

$$\ln |x-1| + \ln |x^2+x+1| = \ln |(x-1)(x^2+x+1)| = \ln |x^3-1| .$$

Solution. 13.m. This is a concise solution written in a form suitable for exam taking. To make this solution as short as possible we have omitted many details. On an exam, the student would be expected to carry out those omitted computations on the side. We set up the partial fraction decomposition as follows.

$$\frac{3x^2+2x-1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1} .$$

Therefore $3x^2+2x-1 = A(x^2+1) + (Bx+C)(x-1)$.

- We set $x = 1$ to get $4 = 2A$, so $A = 2$.
- We set $x = 0$ to get $-1 = A - C$, so $C = 3$.
- Finally, set $x = 2$ to get $15 = 5A + 2B + C$, so $B = 1$.

We can now compute the integral as follows.

$$\int \left(\frac{2}{x-1} + \frac{x+3}{x^2+1} \right) dx = 2 \ln(|x-1|) + \frac{1}{2} \ln(x^2+1) + 3 \arctan x + K .$$

14. Integrate

$$\int \frac{x^6 - x^5 + \frac{9}{2}x^4 - 4x^3 + \frac{13}{2}x^2 - \frac{7}{2}x + \frac{11}{4}}{x^5 - x^4 + 3x^3 - 3x^2 + \frac{9}{4}x - \frac{9}{4}} dx .$$

Solution. 14.

Step 1. The first step of our algorithm is to reduce the fraction so that numerator has smaller degree than the denominator. This is done using polynomial long division as follows.

Variable name(s): x1 division steps total.

	Remainder
	$\frac{3}{2}x^4 - x^3 + \frac{17}{4}x^2 - \frac{5}{4}x + \frac{11}{4}$
Divisor(s)	Quotient(s)
$x^5 - x^4 + 3x^3 - 3x^2 + \frac{9}{4}x - \frac{9}{4}$	x

	Dividend						
-	x^6	$-x^5$	$+\frac{9}{2}x^4$	$-4x^3$	$+\frac{13}{2}x^2$	$-\frac{7}{2}x$	$+\frac{11}{4}$
	x^6	$-x^5$	$+3x^4$	$-3x^3$	$+\frac{9}{4}x^2$	$-\frac{9}{4}x$	
			$\frac{3}{2}x^4$	$-x^3$	$+\frac{17}{4}x^2$	$-\frac{5}{4}x$	$+\frac{11}{4}$

In other words,

$$x^6 - x^5 + \frac{9}{2}x^4 - 4x^3 + \frac{13}{2}x^2 - \frac{7}{2}x + \frac{11}{4} = (x^5 - x^4 + 3x^3 - 3x^2 + \frac{9}{4}x - \frac{9}{4})x + \frac{3}{2}x^4 - x^3 + \frac{17}{4}x^2 - \frac{5}{4}x + \frac{11}{4},$$

and therefore

$$\begin{aligned} \frac{x^6 - x^5 + \frac{9}{2}x^4 - 4x^3 + \frac{13}{2}x^2 - \frac{7}{2}x + \frac{11}{4}}{x^5 - x^4 + 3x^3 - 3x^2 + \frac{9}{4}x - \frac{9}{4}} &= x + \frac{\frac{3}{2}x^4 - x^3 + \frac{17}{4}x^2 - \frac{5}{4}x + \frac{11}{4}}{x^5 - x^4 + 3x^3 - 3x^2 + \frac{9}{4}x - \frac{9}{4}} \\ &= x + \frac{6x^4 - 4x^3 + 17x^2 - 5x + 11}{4x^5 - 4x^4 + 12x^3 - 12x^2 + 9x - 9}. \end{aligned}$$

Set

$$N(x) = 6x^4 - 4x^3 + 17x^2 - 5x + 11$$

and

$$D(x) = 4x^5 - 4x^4 + 12x^3 - 12x^2 + 9x - 9.$$

Step 2. (Split into partial fractions). Factor the denominator $D(x) = 4x^5 - 4x^4 + 12x^3 - 12x^2 + 9x - 9$.

We recall from elementary algebra that there is a trick to find all rational roots of $D(x)$ on condition $D(x)$ has integer coefficients. It is well known that when $\frac{p}{q}$ is a rational number, then $\pm\frac{p}{q}$ may be a root of the integer coefficient polynomial $D(x)$ only if p is a divisor of the constant term of $D(x)$, and q is a divisor of the leading coefficient of $D(x)$. Since in our case the leading coefficient is 4 and the constant term is -9, the only possible rational roots of $D(x)$ are $\pm 1, \pm 3, \pm 9, \pm \frac{1}{2}, \pm \frac{3}{2}, \pm \frac{9}{2}, \pm \frac{1}{4}, \pm \frac{3}{4}, \pm \frac{9}{4}$. A rational number r is a root of $D(x)$ if and only if substituting $x = r$ yields 0. Direct check shows that, for example, $D(-1) = -50$. However, $D(1) = 0$ and therefore using polynomial division we get that $D(x) = (x - 1)(4x^4 + 12x^2 + 9)$. We recognize that the second multiplicand is an exact square and therefore $D(x) = (x - 1)(2x^2 + 3)^2$.

So far we got

$$\frac{N(x)}{D(x)} = \frac{6x^4 - 4x^3 + 17x^2 - 5x + 11}{(x - 1)(2x^2 + 3)^2}.$$

In order to split $\frac{N(x)}{D(x)}$ into partial fractions, we need to find numbers A, B, C, D, E such that

$$\frac{6x^4 - 4x^3 + 17x^2 - 5x + 11}{(x - 1)(2x^2 + 3)^2} = \frac{A}{(x - 1)} + \frac{Bx + C}{(2x^2 + 3)} + \frac{Dx + E}{(2x^2 + 3)^2}.$$

After clearing denominators, we see that this amounts to finding A, B, C, D, E such that

$$6x^4 - 4x^3 + 17x^2 - 5x + 11 = A(2x^2 + 3)^2 + (Bx + C)(2x^2 + 3)(x - 1) + (Dx + E)(x - 1).$$

Plugging in $x = 1$ we see that $25 = 25A$ and so $A = 1$. We may plug back $A = 1$ and regroup to get

$$2x^4 - 4x^3 + 5x^2 - 5x + 2 = (Bx + C)(2x^2 + 3)(x - 1) + (Dx + E)(x - 1).$$

Dividing both sides by $(x - 1)$ we get

$$2x^3 - 2x^2 + 3x - 2 = (Bx + C)(2x^2 + 3) + Dx + E.$$

Regrouping we get

$$x^3(2 - 2B) + x^2(-2 - 2C) + x(3 - 3B - D) + (-2 - 3C - E) = 0.$$

As x is an indeterminate, the above expression may vanish only if all coefficients in the preceding expression vanish. Therefore we get the system

$$\begin{cases} 2 - 2B &= 0 \\ -2 - 2C &= 0 \\ 3 - 3B - D &= 0 \\ -2 - 3C - E &= 0 \end{cases}.$$

We may solve the above linear system using the standard algorithm for solving linear systems (the algorithm is called row reduction and is also known as Gaussian elimination). The latter algorithm is studied in any standard the Linear algebra course. Alternatively, we see from the first equations $B = 1$, $C = -1$, and substituting in the remaining equations we see $D = 0$, $E = 1$. Finally, we check that

$$\frac{x^6 - x^5 + \frac{9}{2}x^4 - 4x^3 + \frac{13}{2}x^2 - \frac{7}{2}x + \frac{11}{4}}{x^5 - x^4 + 3x^3 - 3x^2 + \frac{9}{4}x - \frac{9}{4}} = x + \frac{1}{(x-1)} + \frac{x-1}{(2x^2+3)} + \frac{1}{(2x^2+3)^2} \quad .$$

Step 3. (Find the integral of each partial fraction).

$$\begin{aligned} \int x dx &= \frac{x^2}{2} + C \\ \int \frac{1}{x-1} dx &= \ln|x-1| + C \\ \int \frac{x-1}{2x^2+3} dx &= \int \frac{x}{2x^2+3} dx - \frac{1}{3} \int \frac{1}{\frac{2}{3}x^2+1} dx \\ &= \int \frac{d\left(\frac{x^2}{2}\right)}{2x^2+3} dx - \frac{1}{3} \int \frac{1}{\left(\sqrt{\frac{2}{3}}x\right)^2+1} dx \\ &= \frac{1}{4} \int \frac{d(2x^2+3)}{2x^2+3} dx - \frac{1}{3} \int \frac{\frac{d\left(\sqrt{\frac{2}{3}}x\right)}{\sqrt{\frac{2}{3}}}}{\left(\sqrt{\frac{2}{3}}x\right)^2+1} \\ &= \frac{1}{4} \ln(2x^2+3) - \frac{\sqrt{6}}{6} \arctan\left(\sqrt{\frac{2}{3}}x\right) + C \quad . \end{aligned}$$

The last integral is

$$\begin{aligned} \int \frac{1}{(2x^2+3)^2} dx &= \frac{1}{9} \int \frac{\frac{d\left(\sqrt{\frac{2}{3}}x\right)}{\sqrt{\frac{2}{3}}}}{\left(\left(\sqrt{\frac{2}{3}}x\right)^2+1\right)^2} \\ &= \frac{\sqrt{6}}{18} \int \frac{d\left(\sqrt{\frac{2}{3}}x\right)}{\left(\left(\sqrt{\frac{2}{3}}x\right)^2+1\right)^2} \quad \left| \text{Set } y = \sqrt{\frac{2}{3}}x \right. \\ &= \frac{\sqrt{6}}{18} \int \frac{dy}{(y^2+1)^2} \quad . \end{aligned}$$

The general form of the integral $\int \frac{dy}{(y^2+1)^2}$ is solved in (??) by integration by parts. As a review of the theory, we redo the computations directly.

$$\begin{aligned} C + \arctan y &= \int \frac{dy}{y^2+1} \\ &= \frac{y}{y^2+1} + \int \frac{2y^2 dy}{(y^2+1)^2} = \frac{y}{y^2+1} + \int \frac{2(y^2+1-1)dy}{(y^2+1)^2} \\ &= \frac{y}{y^2+1} + 2 \int \frac{dy}{(y^2+1)} - 2 \int \frac{dy}{(y^2+1)^2} \quad . \end{aligned}$$

Transferring summands we get

$$\int \frac{dy}{(y^2+1)^2} = \frac{1}{2} \left(\frac{y}{y^2+1} + \arctan y \right) + C \quad .$$

We recall that $y = \sqrt{\frac{2}{3}}x$ and therefore

$$\int \frac{dx}{(2x^2+3)^2} = \frac{\sqrt{6}}{36} \left(\frac{\sqrt{\frac{2}{3}}x}{\left(\sqrt{\frac{2}{3}}x\right)^2+1} + \arctan\left(\sqrt{\frac{2}{3}}x\right) \right) + C.$$

To get the final answer we collect all terms:

$$\frac{1}{6} \left(\frac{x}{2x^2+3} \right) - \frac{5\sqrt{6}}{36} \arctan\left(\sqrt{\frac{2}{3}}x\right) + \frac{1}{4} \ln(2x^2+3) + \ln|x-1| + \frac{x^2}{2} + C.$$

15. Integrate.

$$(a) \int \frac{1}{3 + \cos x} dx.$$

$$\frac{1}{2} \ln \left| \frac{\tan \frac{x}{2} + 2}{\tan \frac{x}{2} + \frac{1}{2}} \right| + C$$

$$(b) \int \frac{1}{4 + \cos x} dx.$$

$$\frac{1}{3} \ln \left| \frac{\tan \frac{x}{2} + 3}{\tan \frac{x}{2} + \frac{1}{3}} \right| + C$$

(d) $\int \frac{1}{2 + \tan x} dx$. (Hint: this integral can be done simply with the substitution $x = \arctan t$.)

$$\frac{1}{2} \ln \left| \frac{\tan x + 2}{\tan x + \frac{1}{2}} \right| + C$$

$$(c) \int \frac{1}{3 + \sin x} dx.$$

$$(e) \int \frac{dx}{2 \sin x - \cos x + 5}.$$

$$\frac{1}{5} \ln \left| \frac{\tan \frac{x}{2} + \frac{1}{5}}{\tan \frac{x}{2} + \frac{3}{5}} \right| + C$$

Solution. 15.a We use the standard rationalizing substitution $x = 2 \arctan t$, $t = \tan \left(\frac{x}{2} \right)$. We recall that from the double angle formulas it follows that

$$\cos(2 \arctan t) = \frac{\cos^2(\arctan t) - \sin^2(2 \arctan t)}{\cos^2(\arctan t) + \sin^2(\arctan t)} = \frac{1 - t^2}{1 + t^2}.$$

Therefore we can solve the integral as follows.

$$\begin{aligned} \int \frac{1}{3 + \cos x} dx &= \int \frac{1}{3 + \cos(2 \arctan t)} d(2 \arctan t) \quad \left| \text{Set } x = 2 \arctan t \right. \\ &= \int \frac{1}{\left(3 + \frac{1-t^2}{1+t^2} \right) (1+t^2)} dt \\ &= \int \frac{2}{4 + 2t^2} dt \\ &= \int \frac{1}{2 + t^2} dt \\ &= \frac{\sqrt{2}}{2} \arctan \left(\frac{\sqrt{2}}{2} t \right) + C \\ &= \frac{\sqrt{2}}{2} \arctan \left(\frac{\sqrt{2}}{2} \tan \left(\frac{x}{2} \right) \right) + C. \end{aligned}$$

Solution. 15.d This integral is of none of the forms that can be integrated quickly. Therefore we can solve it using the standard rationalizing substitution $x = 2 \arctan t$, $t = \tan \left(\frac{x}{2} \right)$. This results in somewhat long computations and we invite the reader to try it.

However, as proposed in the hint, the substitution $x = \arctan t$ works much faster:

$$\begin{aligned} \int \frac{1}{2 + \tan x} dx &= \int \frac{1}{2 + \tan(\arctan t)} d(\arctan t) \quad \left| \text{Substitute } x = \arctan t \right. \\ &= \int \frac{1}{(2 + t)(1 + t^2)} dt \quad \left| \text{part. fractions} \right. \\ &= \int \left(\frac{\frac{1}{5}}{(t+2)} + \frac{-\frac{t}{5} + \frac{2}{5}}{(t^2+1)} \right) dt \\ &= \frac{1}{5} \ln |t+2| - \frac{1}{10} \ln(t^2+1) + \frac{2}{5} \arctan t + C \quad \left| t = \tan x \right. \\ &= \frac{1}{5} \ln |\tan x + 2| - \frac{1}{10} \ln(\tan^2 x + 1) + \frac{2}{5} x + C \\ &= \frac{1}{5} \ln |\tan x + 2| + \frac{1}{5} \ln |\cos x| + \frac{2}{5} x + C \\ &= \frac{1}{5} \ln |(\tan x + 2) \cos x| + \frac{2}{5} x + C \\ &= \frac{1}{5} \ln |\sin x + 2 \cos x| + \frac{2}{5} x + C. \end{aligned}$$

Solution. 15.e.

Set $x = 2 \arctan t$. As studied in equation ??, this substitution implies $\cos x = \frac{1-t^2}{1+t^2}$, $\sin x = \frac{2t}{1+t^2}$, $dx = \frac{2}{1+t^2} dt$. Therefore

$$\begin{aligned}
\int \frac{dx}{2 \sin x - \cos x + 5} &= \int \frac{2dt}{(1+t^2) \left(2 \frac{2t}{t^2+1} - \frac{(-t^2+1)}{t^2+1} + 5 \right)} & \left| \begin{array}{l} \text{Set } x = 2 \arctan t \end{array} \right. \\
&= \int \frac{dt}{3t^2 + 2t + 2} \\
&= \int \frac{dt}{3 \left(t^2 + \frac{2}{3}t + \frac{1}{9} - \frac{1}{9} + \frac{2}{3} \right)} \\
&= \int \frac{dt}{3 \left(\left(t + \frac{1}{3} \right)^2 + \frac{5}{9} \right)} \\
&= \int \frac{dt}{\frac{5}{3} \left(\left(\frac{3}{\sqrt{5}} \left(t + \frac{1}{3} \right) \right)^2 + 1 \right)} \\
&= \int \frac{\frac{\sqrt{5}}{3} dw}{\frac{5}{3} (w^2 + 1)} \\
&= \frac{\sqrt{5}}{5} \arctan w + C \\
&= \frac{\sqrt{5}}{5} \arctan \left(\frac{\sqrt{5}}{5} (3t + 1) \right) + C \\
&= \frac{\sqrt{5}}{5} \arctan \left(\frac{\sqrt{5}}{5} \left(3 \tan \left(\frac{x}{2} \right) + 1 \right) \right) + C \quad .
\end{aligned}$$

$$\begin{aligned}
&\text{Set} \\
&w = \frac{3}{\sqrt{5}} \left(t + \frac{1}{3} \right) \\
&= \frac{\sqrt{5}}{5} (3t + 1) \\
&dw = \frac{\sqrt{5}}{5} dt \\
&dt = \frac{\sqrt{5}}{5} dw
\end{aligned}$$

16. Integrate. The answer key has not been proofread, use with caution.

(a) $\int \sin(3x) \cos(2x) dx.$

ANSWER: $-\frac{1}{10} \cos(5x) + \frac{1}{10} \cos(x) + C$

(b) $\int \sin x \cos(5x) dx.$

ANSWER: $-\frac{1}{24} \cos(6x) + \frac{1}{24} \cos(4x) + C$

(c) $\int \cos(3x) \sin(2x) dx.$

ANSWER: $-\frac{1}{10} \cos(5x) + \frac{1}{10} \cos(x) + C$

(d) $\int \sin(5x) \sin(3x) dx.$

ANSWER: $\frac{1}{12} \sin(8x) - \frac{1}{12} \sin(2x) + C$

(e) $\int \cos(x) \cos(3x) dx.$

ANSWER: $\frac{1}{8} \sin(4x) + \frac{1}{8} \sin(2x) + C$

17. Integrate.

(a) $\int \sin^2 x \cos x dx.$

ANSWER: $\frac{1}{3} \sin^3 x + C$

(c) $\int \cos^3 x dx.$

ANSWER: $\sin x - \frac{1}{3} \sin^3 x + C$

(b) $\int \sin^2 x dx.$

ANSWER: $\frac{1}{2} \sin(2x) - \frac{\pi}{2} + C$

(d) $\int \sin^3 x \cos^4 x dx.$

ANSWER: $\frac{1}{5} \cos^5 x - \frac{1}{5} \cos^3 x + C$

18. Integrate.

(a) $\int \sec x dx.$

ANSWER: $\ln \left| \sec x + \tan x \right| = \ln \left| \frac{1 + \tan \left(\frac{x}{2} \right)}{1 - \tan \left(\frac{x}{2} \right)} \right| + C$

(b) $\int \sec^3 x dx.$

ANSWER: $\frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + C$

$$(c) \int \tan^3 x dx.$$

$$(d) \int \sec^2 x \tan^2 x dx.$$

$$C + |x \sec x| \ln |x| - x \tan^2 \frac{x}{2} + C$$

$$C + \frac{6}{x} \tan^2 \frac{x}{2} + C$$

Solution. 18.a. Variant I.

This variant uses the standard method for solving trigonometric integrals with the substitution $x = \arctan(2t)$.

$$\begin{aligned} \int \sec x dx &= \int \sec(2 \arctan t) d(2 \arctan t) && \left| \begin{array}{l} \text{Set } x = 2 \arctan t \\ \text{Use } \cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z} \end{array} \right. \\ &= \int \frac{1}{\cos(2 \arctan t)} \frac{2}{1 + t^2} dt \\ &= \int \frac{1}{\frac{1 - t^2}{1 + t^2}} \frac{2}{1 + t^2} dt \\ &= \int \frac{2}{1 - t^2} dt && \left| \begin{array}{l} \text{part. fractions} \end{array} \right. \\ &= \int \left(\frac{1}{1 - t} + \frac{1}{1 + t} \right) dt \\ &= -\ln |1 - t| + \ln |1 + t| + C \\ &= \ln \left| \frac{1 + t}{1 - t} \right| && \left| \begin{array}{l} \text{Subst. } t = \tan \left(\frac{x}{2} \right) \end{array} \right. \\ &= \ln \left| \frac{1 + \tan \left(\frac{x}{2} \right)}{1 - \tan \left(\frac{x}{2} \right)} \right| + C && \left| \begin{array}{l} \text{Last step: see below} \end{array} \right. \\ &= \ln |\sec x + \tan x| + C. \end{aligned}$$

The expression $\ln \left| \frac{1 + \tan \left(\frac{x}{2} \right)}{1 - \tan \left(\frac{x}{2} \right)} \right|$ presents a perfectly good answer, which would certainly would qualify for a correct test answer.

However, as shown above, it can be rewritten into the shorter form $\ln |\sec x + \tan x|$. Below we quickly prove that $\frac{1 + \tan \left(\frac{x}{2} \right)}{1 - \tan \left(\frac{x}{2} \right)}$ equals $\sec x + \tan x$.

$$\begin{aligned} \sec x + \tan x &= \frac{1 + \sin x}{\cos x} && \left| \begin{array}{l} \text{Use:} \\ \sin x = 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right) \\ \cos x = \cos^2 \left(\frac{x}{2} \right) - \sin^2 \left(\frac{x}{2} \right) \\ 1 = \cos^2 \left(\frac{x}{2} \right) + \sin^2 \left(\frac{x}{2} \right) \end{array} \right. \\ &= \frac{\cos^2 \left(\frac{x}{2} \right) + \sin^2 \left(\frac{x}{2} \right) + 2 \sin \left(\frac{x}{2} \right) \cos \left(\frac{x}{2} \right)}{\cos^2 \left(\frac{x}{2} \right) - \sin^2 \left(\frac{x}{2} \right)} \\ &= \frac{(\sin \left(\frac{x}{2} \right) + \cos \left(\frac{x}{2} \right))^2}{(\cos \left(\frac{x}{2} \right) - \sin \left(\frac{x}{2} \right)) (\cos \left(\frac{x}{2} \right) + \sin \left(\frac{x}{2} \right))} \\ &= \frac{(\cos \left(\frac{x}{2} \right) - \sin \left(\frac{x}{2} \right)) \frac{1}{\cos \left(\frac{x}{2} \right)}}{(\cos \left(\frac{x}{2} \right) - \sin \left(\frac{x}{2} \right)) \frac{1}{\cos \left(\frac{x}{2} \right)}} \\ &= \frac{1 + \tan \left(\frac{x}{2} \right)}{1 - \tan \left(\frac{x}{2} \right)} \end{aligned}$$

18.a. **Variant II.** This variant is based on the following observation. For an odd number $m > 0$, we studied a quick technique for integrating $\int \sin^n x \cos^m x dx$: namely, use the transformation $\cos x dx = d(\sin x)$ and change variables $u = \sin x$. This trick relies heavily on the fact that m is odd (as we need to express the remaining even power of $\cos x$ via $\sin x$). However, the positivity of m is not essential: by multiplying top and bottom by $\cos x$ we can make this technique work also for odd negative values of m . We illustrate the technique in the solution below.

$$\begin{aligned}
\int \sec x dx &= \int \frac{1}{\cos x} dx \\
&= \int \frac{\cos x}{\cos^2 x} dx \\
&= \int \frac{d(\sin x)}{1 - \sin^2 x} && \left| \text{Set } u = \sin x \right. \\
&= \int \frac{du}{1 - u^2} \\
&= \int \frac{du}{(1+u)(1-u)} && \left| \text{part. fractions} \right. \\
&= \int \left(\frac{\frac{1}{2}}{1+u} + \frac{\frac{1}{2}}{1-u} \right) du \\
&= \frac{1}{2} (\ln |1+u| - \ln |1-u|) + C \\
&= \frac{1}{2} \ln \left| \frac{1+u}{1-u} \right| + C && \left| \text{Subst. back } u = \sin x \right. \\
&= \frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C && \left| \text{Last step: see below} \right. \\
&= \ln |\sec x + \tan x| + C .
\end{aligned}$$

The expression $\frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C$ gives a perfectly good answer (which may be the preferred answer depending on the textbook). Let us show however that $\frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right|$ equals $\ln |\sec x + \tan x|$, the answer given in the other variants.

$$\begin{aligned}
\frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| &= \frac{1}{2} \ln \left| \frac{1+\sin x}{1-\sin x} \right| && \left| \text{Mult. \& div by } 1+\sin x \right. \\
&= \frac{1}{2} \ln \left| \frac{(1+\sin x)^2}{(1-\sin x)(1+\sin x)} \right| \\
&= \frac{1}{2} \ln \left| \frac{(1+\sin x)^2}{\cos^2 x} \right| && \left| \text{use } \frac{1}{2} \ln |a| = \ln |a|^{\frac{1}{2}} \right. \\
&= \ln \sqrt{\left| \frac{(1+\sin x)^2}{\cos^2 x} \right|} \\
&= \ln \left| \frac{1+\sin x}{\cos x} \right| \\
&= \ln |\sec x + \tan x| .
\end{aligned}$$

18.a. Variant III. This variant present a quick solution by multiplying and dividing our integrand by the multiplier $\sec x + \tan x$. Of course, the idea of using that multiplier comes from knowing the answer to the problem in advance (which can be obtained, for example, by using the preceding solution variants).

$$\begin{aligned}
\int \sec x dx &= \int \sec x \frac{\sec x + \tan x}{\sec x + \tan x} dx \\
&= \int \frac{\sec^2 x + \sec x \tan x}{\sec x + \tan x} dx && \left| \begin{array}{l} d(\tan x) = \sec^2 x dx \\ d(\sec x) = \sec x \tan x dx \end{array} \right. \\
&= \int \frac{d(\sec x + \tan x)}{\sec x + \tan x} && \left| \text{Set } u = \sec x + \tan x \right. \\
&= \int \frac{du}{u} \\
&= \ln |u| + C \\
&= \ln |\sec x + \tan x| + C .
\end{aligned}$$

Solution. 18.b This problem can be solved with the general method by setting $x = 2 \arctan t$. However, there are shorter ways to solve the integral, as we show below.

Variant I.

$$\begin{aligned}
\int \sec^3 x dx &= \int \frac{1}{\cos^3 x} dx \\
&= \int \frac{\cos x}{\cos^4 x} dx && \text{use } d(\sin x) = \cos x dx \\
&= \int \frac{1}{\cos^4 x} d(\sin x) && \text{use } \cos^2 x = 1 - \sin^2 x \\
&= \int \frac{1}{(1 - \sin^2 x)^2} d(\sin x) && \text{Set } \sin x = u \\
&= \int \frac{1}{(1 - u^2)^2} du && \text{split in part. frac.} \\
&= \int \left(\frac{\frac{1}{4}}{u+1} + \frac{\frac{1}{4}}{(u+1)^2} + \frac{-\frac{1}{4}}{u-1} + \frac{\frac{1}{4}}{(u-1)^2} \right) du \\
&= \frac{1}{4} \left(\ln |u+1| - \ln |u-1| - \frac{1}{u+1} - \frac{1}{u-1} \right) + C \\
&= \frac{1}{4} \left(\ln \left| \frac{u+1}{u-1} \right| - \frac{2u}{u^2-1} \right) + C \\
&= \frac{1}{4} \left(\ln \left| \frac{\sin x + 1}{\sin x - 1} \right| + \frac{2 \sin x}{\cos^2 x} \right) + C.
\end{aligned}$$

Variation II. This variation uses the preceding problem to get to a solution as follows.

$$\begin{aligned}
\int \sec^3 x dx &= \int \sec x d(\tan x) && \text{int. by parts} \\
&= \sec x \tan x - \int \tan x d(\sec x) \\
&= \sec x \tan x - \int \sec x \tan^2 x dx && \tan^2 x = \sec^2 x - 1 \\
&= \sec x \tan x - \int \sec x (\sec^2 x - 1) dx \\
&= \sec x \tan x - \int \sec^3 x dx + \int \sec x dx && \text{Use Problem 18.a} \\
&= \sec x \tan x - \int \sec^3 x dx + \ln |\sec x + \tan x| && + \int \sec^3 x dx \\
&&& \text{to both sides} \\
2 \int \sec^3 x dx &= (\sec x \tan x + \ln |\sec x + \tan x|) + C \\
\int \sec^3 x dx &= \frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + K.
\end{aligned}$$

19. Find a linear substitution (via completing the square) to transform the radical to a multiple of an expression of the form $\sqrt{u^2 + 1}$, $\sqrt{u^2 - 1}$ or $\sqrt{1 - u^2}$.

- (a) $\sqrt{x^2 + x + 1}$.
(b) $\sqrt{-2x^2 + x + 1}$.

Solution. 19.a

$$\begin{aligned}
\sqrt{x^2 + x + 1} &= \sqrt{x^2 + 2\frac{1}{2}x + \frac{1}{4} - \frac{1}{4} + 1} \\
&= \sqrt{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} \\
&= \sqrt{\frac{3}{4} \left(\frac{4}{3} \left(x + \frac{1}{2}\right)^2 + 1\right)} \\
&= \frac{\sqrt{3}}{2} \sqrt{\left(\frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right)\right)^2 + 1} \\
&= \frac{\sqrt{3}}{2} \sqrt{u^2 + 1},
\end{aligned}$$

where $u = \frac{2}{\sqrt{3}} \left(x + \frac{1}{2}\right) = \frac{2\sqrt{3}}{3}x + \frac{\sqrt{3}}{3}$.

Solution. 19.b

$$\begin{aligned}
 \sqrt{-2x^2 + x + 1} &= \sqrt{-2 \left(x^2 - \frac{1}{2}x - \frac{1}{2} \right)} \\
 &= \sqrt{-2 \left(x^2 - 2\frac{1}{4}x + \frac{1}{16} - \frac{1}{16} - \frac{1}{2} \right)} \\
 &= \sqrt{-2 \left(\left(x - \frac{1}{16} \right)^2 - \frac{9}{16} \right)} \\
 &= \sqrt{\frac{9}{8} \left(-\frac{16}{9} \left(x - \frac{1}{16} \right)^2 + 1 \right)} \\
 &= \frac{3}{\sqrt{8}} \sqrt{-\left(\frac{4}{3} \left(x - \frac{1}{16} \right) \right)^2 + 1} \\
 &= \frac{3}{\sqrt{8}} \sqrt{-u^2 + 1}
 \end{aligned}$$

where $u = \frac{4}{3} \left(x - \frac{1}{16} \right) = \frac{4}{3}x - \frac{1}{12}$.

20. Compute the integral.

(a) $\int \frac{\sqrt{1+x^2}}{x^2} dx.$

$$-\frac{x}{\sqrt{1+x^2}} - \ln \left(x + \sqrt{1+x^2} \right) + C$$

Solution. 20.a

Variant I. In this variant, we use the trigonometric substitution $x = \tan \theta$ and then solve the integral using a few algebraic tricks.

$$\begin{aligned} \int \frac{\sqrt{1+x^2}}{x^2} dx &= \int \frac{\sqrt{1+\tan^2 \theta}}{\tan^2 \theta} d(\tan \theta) \\ &= \int \frac{|\sec \theta|}{\tan^2 \theta} \sec^2 \theta d\theta \\ &= \int \frac{\cos^2 \theta}{\cos^3 \theta \sin^2 \theta} d\theta \\ &= \int \frac{\cos \theta}{\cos^2 \theta \sin^2 \theta} d\theta \end{aligned}$$

$$\begin{aligned} \text{Set} \\ x &= \tan \theta \\ \theta &\in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \\ |\sec \theta| &= \sec \theta \\ \text{for } \theta &\in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \end{aligned}$$

$$= \int \frac{d(\sin \theta)}{(1 - \sin^2 \theta) \sin^2 \theta}$$

$$\begin{aligned} \text{Set} \\ u &= \sin \theta \\ \text{for } \theta &\in \left(0, \frac{\pi}{2}\right) \\ u &= \sqrt{1 - \cos^2 \theta} \\ u &= \sqrt{1 - \frac{1}{\sec^2 \theta}} \\ u &= \sqrt{1 - \frac{1}{1 + \tan^2 \theta}} \\ u &= \sqrt{\frac{\tan^2 \theta}{1 + \tan^2 \theta}} \\ u &= \frac{\tan \theta}{\sqrt{1 + \tan^2 \theta}} \\ u &= \frac{x}{\sqrt{1 + x^2}} \end{aligned}$$

$$\begin{aligned} &= \int \frac{du}{(1 - u^2)u^2} \\ &= \int \frac{du}{(1 - u)u^2(u + 1)} \\ &= \int \left(\frac{\frac{1}{2}}{u + 1} + \frac{-\frac{1}{2}}{u - 1} + \frac{1}{u^2} \right) du \\ &= -\frac{1}{2} \ln |u - 1| + \frac{1}{2} \ln (u + 1) - u^{-1} + C \\ &= -\frac{1}{2} \ln (1 - u) + \frac{1}{2} \ln (u + 1) - u^{-1} + C \\ &= \frac{1}{2} \ln \left(\frac{1 + u}{1 - u} \right) - u^{-1} + C \\ &= \frac{1}{2} \ln \left(\frac{(1 + u)}{(1 - u)} \cdot \frac{(1 + u)}{(1 + u)} \right) - u^{-1} + C \\ &= \frac{1}{2} \ln \left(\frac{(1 + u)^2}{1 - u^2} \right) - u^{-1} + C \\ &= \frac{1}{2} \ln \left(\frac{(1 + u)^2}{\frac{1}{1 + x^2}} \right) - \frac{\sqrt{1 + x^2}}{x} + C \\ &= \frac{1}{2} \ln \left(\left((1 + u) \sqrt{1 + x^2} \right)^2 \right) - \frac{\sqrt{1 + x^2}}{x} + C \\ &= \ln \left(\sqrt{1 + x^2} + x \right) - \frac{\sqrt{1 + x^2}}{x} + C \quad . \end{aligned}$$

use part. frac.

$$u = \frac{x}{\sqrt{1 + x^2}} < 1$$

$$\text{use } u = \frac{x}{\sqrt{1 + x^2}}$$

Variant II. In this variant, we use directly the Euler substitution

$$\begin{aligned}
x &= \cot(2 \arctan t) \\
&= \frac{1}{2} \left(\frac{1}{t} - t \right) \\
dx &= -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\
\sqrt{1+x^2} &= \frac{1}{2} \left(\frac{1}{t} + t \right) \\
t &= \sqrt{x^2+1} - x \\
\frac{1}{t} &= \sqrt{x^2+1} + x \quad .
\end{aligned}$$

$$\begin{aligned}
\int \frac{\sqrt{1+x^2}}{x^2} dx &= \int \frac{\frac{1}{2} \left(\frac{1}{t} + t \right)}{\frac{1}{4} \left(\frac{1}{t} - t \right)^2} \left(-\frac{1}{2} \right) \left(\frac{1}{t^2} + 1 \right) dt \\
&= \int \frac{-t^4 - 2t^2 - 1}{(t-1)^2 t (t+1)^2} dt && \left| \begin{array}{l} \text{Part. frac} \end{array} \right. \\
&= \int \left(-\frac{1}{t} + \frac{1}{(t+1)^2} - \frac{1}{(t-1)^2} \right) dt \\
&= -\ln t - \frac{1}{t+1} + \frac{1}{t-1} + C \\
&= \ln \left(\frac{1}{t} \right) + \frac{2}{t^2-1} + C \\
&= \ln \left(\sqrt{1+x^2} + x \right) + \frac{1}{t \cdot \frac{1}{2} \left(\frac{1}{t} - t \right)} + C \\
&= \ln \left(\sqrt{1+x^2} + x \right) - \frac{1}{t} \cdot \frac{1}{\frac{1}{2} \left(\frac{1}{t} - t \right)} + C \\
&= \ln \left(\sqrt{1+x^2} + x \right) - \left(\sqrt{x^2+1} + x \right) \cdot \frac{1}{x} + C \\
&= \ln \left(\sqrt{1+x^2} + x \right) - \frac{\sqrt{x^2+1}}{x} - 1 + C \quad .
\end{aligned}$$

21. Compute the integral using a trigonometric substitution.

$$(a) \int \frac{\sqrt{9-x^2}}{x^2} dx \quad .$$

$$C + \left(\frac{3}{x} \right) - \frac{x}{\sqrt{9-x^2}} - \arcsin \frac{x}{3} \quad \text{ANSWER:}$$

Solution. 21.a

$$\begin{aligned}
\int \frac{\sqrt{9-x^2}}{x^2} dx &= \int \frac{3\sqrt{\cos^2 \theta}}{9 \sin^2 \theta} (3 \cos \theta) d\theta && \left| \begin{array}{l} \text{Set } x = 3 \sin \theta \\ \text{for } \theta \in \left[\frac{\pi}{2}, 0 \right) \cup \left(0, \frac{\pi}{2} \right] \\ dx = 3 \cos \theta d\theta \\ \text{For } \theta \in \left[\frac{\pi}{2}, 0 \right) \cup \left(0, \frac{\pi}{2} \right] \\ \text{we have } |\cos \theta| = \cos \theta \end{array} \right. \\
&= 9 \int \frac{|\cos \theta|}{\sin^2 \theta} \cos \theta d\theta \\
&= \int \cot^2 \theta d\theta \\
&= \int (\csc^2 \theta - 1) d\theta \\
&= -\cot \theta - \theta + C \\
&= -\frac{\sqrt{9-x^2}}{x} - \arcsin \left(\frac{x}{3} \right) + C,
\end{aligned}$$

where we expressed $\cot \theta$ via $\sin \theta$ by considering the following triangle.

22. Compute the integral.

$$(a) \int \sqrt{x^2+1} dx$$

$$C + \left(x + \sqrt{x^2+1} \right) \ln \left| \frac{x}{\sqrt{x^2+1}} + 1 + \sqrt{x^2+1} \right| \quad \text{ANSWER:}$$

$$(b) \int \sqrt{x^2+2} dx$$

$$C + \sqrt{x^2+2} \ln \left| \frac{x}{\sqrt{x^2+2}} + 1 + \sqrt{x^2+2} \right| + \left(\frac{x}{\sqrt{x^2+2}} + 1 + \sqrt{x^2+2} \right) \ln \left| \frac{x}{\sqrt{x^2+2}} + 1 + \sqrt{x^2+2} \right| \quad \text{ANSWER:}$$

$$(c) \int \sqrt{x^2+x+1} dx$$

$$C + \left(1 + \sqrt{\left(\frac{x}{4} + x \right) \left(\frac{x}{4} + x \right) + 1} \right) \ln \left| \frac{x}{\sqrt{\left(\frac{x}{4} + x \right) \left(\frac{x}{4} + x \right) + 1} + 1 + \sqrt{\left(\frac{x}{4} + x \right) \left(\frac{x}{4} + x \right) + 1}} \right| + \left(\left(\frac{x}{4} + x \right) \ln \left(\frac{x}{4} + x \right) + \sqrt{\left(\frac{x}{4} + x \right) \left(\frac{x}{4} + x \right) + 1} \right) \ln \left| \frac{x}{\sqrt{\left(\frac{x}{4} + x \right) \left(\frac{x}{4} + x \right) + 1} + 1 + \sqrt{\left(\frac{x}{4} + x \right) \left(\frac{x}{4} + x \right) + 1}} \right| \quad \text{ANSWER:}$$

$$(d) \int \sqrt{(2x^2 + 2x + 1)} dx$$

$$(e) \int \sqrt{(3x^2 + 2x + 1)} dx$$

$$(f) \int \frac{\sqrt{x^2 + 1}}{x + 1} dx$$

Solution. 22.a.

This problem can be solved both via the Euler substitution and by transforming to a trigonometric integral and solving the trigonometric integral on its own. We present both variants.

Variant I. We recall the Euler substitution for $\sqrt{x^2 + 1}$ given in (??):

$$\begin{aligned} x &= \frac{1}{2} \left(\frac{1}{t} - t \right) \\ \sqrt{x^2 + 1} &= \frac{1}{2} \left(\frac{1}{t} + t \right) \\ dx &= -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\ t &= \sqrt{x^2 + 1} - x \end{aligned}$$

Therefore

$$\begin{aligned} \int \sqrt{x^2 + 1} dx &= -\int \frac{1}{4} \left(\frac{1}{t} + t \right) \left(\frac{1}{t^2} + 1 \right) dt \\ &= -\frac{1}{4} \int \left(\frac{1}{t^3} + 2\frac{1}{t} + t \right) dt \\ &= -\frac{1}{4} \left(-\frac{t^{-2}}{2} + 2 \ln |t| + \frac{t^2}{2} \right) + C \\ &= \frac{1}{8} (t^{-2} - t^2) - \frac{1}{2} \ln |t| + C \\ &= \frac{1}{2} \left(\underbrace{\frac{1}{2} (t^{-1} - t)}_{=x} \right) \left(\underbrace{\frac{1}{2} (t^{-1} + t)}_{=\sqrt{x^2+1}} \right) - \frac{1}{2} \ln |t| + C \\ &= \frac{1}{2} x \sqrt{x^2 + 1} - \frac{1}{2} \ln |\sqrt{x^2 + 1} - x| + C \\ &= \frac{1}{2} x \sqrt{x^2 + 1} + \frac{1}{2} \ln (\sqrt{x^2 + 1} + x) + C \end{aligned}$$

$\left. \begin{array}{l} a^2 - b^2 = \\ (a - b)(a + b) \end{array} \right|$
 $\left. \begin{array}{l} \text{See below} \end{array} \right|$

Our problem is solved.

A few comments are in order. In the above expression we would have obtained a perfectly good answer if we plugged in $t = \sqrt{x^2 + 1} - x$ into the fourth line, however our answer would look much more complicated. Indeed, had we not used the formula $a^2 - b^2 = (a - b)(a + b)$ in the fourth line, the term $t^{-2} - t^2$ would be equal to $\frac{1}{(\sqrt{x^2+1}-x)^2} - (\sqrt{x^2+1}-x)^2$. In turn, the term $\frac{1}{(\sqrt{x^2+1}-x)^2} - (\sqrt{x^2+1}-x)^2$ can be simplified to $4x\sqrt{x^2+1}$ as follows. We carry out the simplifications to illustrate some of the algebraic issues arising when dealing with integrals of radicals.

$$\begin{aligned} t^{-2} - t^2 &= \frac{1}{(\sqrt{x^2+1}-x)^2} - (\sqrt{x^2+1}-x)^2 \\ &= \frac{(\sqrt{x^2+1}+x)^2}{(\sqrt{x^2+1}-x)^2(\sqrt{x^2+1}+x)^2} \\ &\quad - (\sqrt{x^2+1}-x)^2 \\ &= \frac{(\sqrt{x^2+1}+x)^2}{\underbrace{((\sqrt{x^2+1})^2 - x^2)^2}_{=1}} - (\sqrt{x^2+1}-x)^2 \\ &= 4x\sqrt{x^2+1} \end{aligned}$$

Of course, the above computations are unnecessary if we use the formula $a^2 - b^2 = (a - b)(a + b)$ as done in the original solution.

We note that in the last transformation we transformed $\ln |\sqrt{x^2 + 1} - x|$ to $\ln (\sqrt{x^2 + 1} - x)$ because the quantity $\sqrt{x^2 + 1} - x$ is always positive. The proof of that fact we leave for the reader's exercise.

Finally, we note that as a last simplification to our solution, we used the transformation $\ln |t| = \ln (\sqrt{x^2 + 1} - x) = -\ln |\frac{1}{t}| = -\ln (\sqrt{x^2 + 1} + x)$. This is seen as follows.

$$\begin{aligned}
 \ln |t| &= -\ln \left| \frac{1}{t} \right| \\
 &= -\ln \left(\frac{1}{\sqrt{x^2 + 1} - x} \right) && \left| \begin{array}{l} \text{rationalize} \end{array} \right. \\
 &= -\ln \left(\frac{(\sqrt{x^2 + 1} + x)}{(\sqrt{x^2 + 1} - x)(\sqrt{x^2 + 1} + x)} \right) \\
 &= -\ln \left(\frac{\sqrt{x^2 + 1} + x}{x^2 + 1 - x^2} \right) \\
 &= -\ln (\sqrt{x^2 + 1} + x) \quad .
 \end{aligned}$$

Variant II. In this variant we transform to a trigonometric integral and solve it using ad-hoc methods. We recall that if we decided to solve the trigonometric integral using the standard substitution $\theta = 2 \arctan t$, we would arrive at the Euler substitution given in Variant I.

$ \begin{aligned} \int \sqrt{x^2 + 1} dx &= \int \sqrt{\tan^2 \theta + 1} d(\tan \theta) \\ &= \int \sqrt{\sec^2 \theta} \sec^2 \theta d\theta \\ &= \int \sec^3 \theta d\theta \\ &= \frac{1}{2} (\tan \theta \sec \theta + \ln \sec \theta + \tan \theta) + C \\ &= \frac{1}{2} \left(x \sqrt{x^2 + 1} + \ln (\sqrt{x^2 + 1} + x) \right) + C \end{aligned} $	Set $x = \tan \theta$ $\theta \in (-\frac{\pi}{2}, \frac{\pi}{2})$ $\sec \theta > 0$ Problem 18.b $\sec \theta = \sqrt{x^2 + 1}$ $\tan \theta = x$
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Solution. 22.d

$$\begin{aligned}
 \int \sqrt{(2x^2 + 2x + 1)} dx &= \int \sqrt{2} \sqrt{\left(\left(x + \frac{1}{2} \right)^2 + \frac{1}{4} \right)} dx && \left| \begin{array}{l} \text{complete square} \end{array} \right. \\
 &= \sqrt{2} \int \sqrt{\frac{1}{4} \left(4 \left(x + \frac{1}{2} \right)^2 + 1 \right)} dx \\
 &= \frac{\sqrt{2}}{2} \int \sqrt{\left(4 \left(x + \frac{1}{2} \right)^2 + 1 \right)} dx \\
 &= \frac{\sqrt{2}}{2} \int \sqrt{\left((2x + 1)^2 + 1 \right)} \frac{1}{2} d(2x + 1) && \left| \begin{array}{l} \text{Set } u = 2x + 1 \\ \text{Euler subst.:} \\ u = \frac{1}{2} \left(\frac{1}{t} - t \right), \\ t > 0 \\ \\ du = -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\ \sqrt{u^2 + 1} = \frac{1}{2} \left(\frac{1}{t} + t \right) \\ t = \sqrt{u^2 + 1} - u \end{array} \right. \\
 &= \frac{\sqrt{2}}{4} \int \sqrt{(u^2 + 1)} du \\
 &= -\frac{\sqrt{2}}{16} \int \left(\frac{1}{t} + t \right) \left(\frac{1}{t^2} + 1 \right) dt \\
 &= -\frac{\sqrt{2}}{16} \int (t^{-3} + 2t^{-1} + t) dt \\
 &= -\frac{\sqrt{2}}{16} \left(-\frac{t^{-2}}{2} + 2 \ln |t| + \frac{t^2}{2} \right) + C && \left| \begin{array}{l} \text{simplify as} \\ \text{in Problem 22.a} \end{array} \right. \\
 &= \frac{\sqrt{2}}{8} \left(u \sqrt{u^2 + 1} + \ln \left(\sqrt{u^2 + 1} + u \right) \right) + C \\
 &= \frac{\sqrt{2}}{8} \left((2x + 1) \sqrt{(2x + 1)^2 + 1} \right. \\
 &\quad \left. + \ln \left(\sqrt{(2x + 1)^2 + 1} + 2x + 1 \right) \right) + C.
 \end{aligned}$$

Solution. 22.f

$$\begin{aligned}
\int \frac{\sqrt{x^2+1}}{x+1} dx &= \int \frac{\frac{1}{2} \left(\frac{1}{t} + t \right)}{\frac{1}{2} \left(\frac{1}{t} - t \right) + 1} d \left(\frac{1}{2} \left(\frac{1}{t} - t \right) \right) && \left| \begin{array}{l} \text{Euler sub:} \\ x = \frac{1}{2} \left(\frac{1}{t} - t \right) \\ \sqrt{x^2+1} = \frac{1}{2} \left(\frac{1}{t} + t \right) \end{array} \right. \\
&= \int \left(\frac{1+t^2}{1-t^2+2t} \right) \frac{1}{2} (-t^{-2} - 1) dt \\
&= \int \frac{1}{2} \frac{(1+t^2)(-t^{-2}-1)}{1-t^2+2t} dt \\
&= \frac{1}{2} \int \frac{t^4 + 2t^2 + 1}{t^4 - 2t^3 - t^2} dt && \left| \begin{array}{l} \text{pol. long div.} \\ \text{part. fractions} \end{array} \right. \\
&= \frac{1}{2} \int \left(1 + \frac{2t^3 + 3t^2 + 1}{t^2(t^2 - 2t - 1)} \right) dt \\
&= \frac{1}{2} \int \left(1 + \frac{2\sqrt{2}}{t - \sqrt{2} - 1} + \frac{-2\sqrt{2}}{t + \sqrt{2} - 1} + \frac{2}{t} + \frac{-1}{t^2} \right) dt \\
&= -\sqrt{2} \ln |t + \sqrt{2} - 1| + \sqrt{2} \ln |t - \sqrt{2} - 1| \\
&\quad + \frac{1}{2} t^{-1} + \ln |t| + \frac{1}{2} t + C && \left| t = \sqrt{x^2+1} - x \right. \\
&= -\sqrt{2} \ln \left(\sqrt{x^2+1} - x + \sqrt{2} - 1 \right) \\
&\quad + \sqrt{2} \ln \left(\sqrt{x^2+1} - x - \sqrt{2} - 1 \right) \\
&\quad + \ln \left(\sqrt{x^2+1} - x \right) \\
&\quad + \frac{1}{2} \left(\sqrt{x^2+1} - x \right)^{-1} + \frac{1}{2} \sqrt{x^2+1} - \frac{1}{2} x + C && \left| \begin{array}{l} \text{Last 3 terms} \\ \text{simplify} \end{array} \right. \\
&= -\sqrt{2} \ln \left(\sqrt{x^2+1} - x + \sqrt{2} - 1 \right) \\
&\quad + \sqrt{2} \ln \left(\sqrt{x^2+1} - x - \sqrt{2} - 1 \right) \\
&\quad + \ln \left(\sqrt{x^2+1} - x \right) \\
&\quad + \sqrt{x^2+1} + C .
\end{aligned}$$

23. Let $b^2 - 4ac < 0$ and $a > 0$ be (real) numbers. Show that

$$\int \sqrt{ax^2 + bx + c} dx = \frac{\sqrt{a}D}{2} \left(\ln \left(\sqrt{\left(\frac{2xa+b}{2\sqrt{D}a} \right)^2 + 1} + \frac{2xa+b}{2\sqrt{D}a} \right) + \frac{2xa+b}{2\sqrt{D}a} \sqrt{\left(\frac{2xa+b}{2\sqrt{D}a} \right)^2 + 1} \right) + C,$$

$$\text{where } D = \frac{4ac - b^2}{4a^2}.$$

24. Integrate

(a) $\int \sqrt{1-x^2} dx$

(b) $\int \sqrt{2-x^2} dx$

(c) $\int \sqrt{-x^2 + x + 1} dx$

(d) $\int \sqrt{2-x-x^2} dx$

(e) $\int \frac{\sqrt{1-x^2}}{1+x} dx$

(f) $\int \frac{\sqrt{1-x^2}}{2+x} dx$

Solution. 24.a

Variant I. This integral can quickly be solved using a trig substitution. The Euler substitution results in a slightly longer solution, shown in the next solution variant.

$$\begin{aligned}
\int \sqrt{1-x^2} dx &= \int \sqrt{1-\cos^2 \theta} d(\cos \theta) && \left| \begin{array}{l} \text{Set } x = \cos \theta, \theta \in [0, \pi] \\ \theta \in [0, \pi] \Rightarrow \sin \theta \geq 0 \\ \sin^2 \theta = \frac{1-\cos(2\theta)}{2} \end{array} \right. \\
&= \int \sqrt{\sin^2 \theta} (-\sin \theta) d\theta \\
&= -\int \sin^2 \theta d\theta \\
&= -\int \frac{1-\cos(2\theta)}{2} d\theta \\
&= -\frac{\theta}{2} + \frac{\sin(2\theta)}{4} + C \\
&= -\frac{\theta}{2} + \frac{2 \sin \theta \cos \theta}{4} + C && \left| \begin{array}{l} x = \cos \theta \\ \theta = \arccos x \\ \sin \theta = \sin(\arccos x) \\ = \sqrt{1-x^2} \end{array} \right. \\
&= -\frac{\arccos x}{2} + \frac{x\sqrt{1-x^2}}{2} + C \\
&= \frac{\arcsin x}{2} + \frac{x\sqrt{1-x^2}}{2} + K,
\end{aligned}$$

where for the last equality we recall that the

derivative of $\arcsin x$ is minus the derivative of $\arccos x$.

Variant II. We show how to do this integral via the Euler substitution $x = \cos(2 \arctan t)$.

$$\begin{aligned}
\int \sqrt{1-x^2} dx &= \int \sqrt{1-\cos^2 \theta} d(\cos \theta) && \left| \begin{array}{l} \text{Set} \\ x = \cos(2 \arctan t) \\ \frac{1}{2} \arccos x = \arctan t \\ x = \frac{1-t^2}{1+t^2} \\ = \frac{2}{1+t^2} - 1 \\ \sqrt{1-x^2} = \frac{2t}{1+t^2} \end{array} \right. \\
&= \int \frac{2t}{1+t^2} d\left(\frac{1-t^2}{1+t^2}\right) \\
&= \int \frac{2t}{1+t^2} \left(\frac{-4t}{(1+t^2)^2}\right) dt && \left| \begin{array}{l} \text{Integral rational} \\ \text{function} \\ \text{we skip details} \end{array} \right. \\
&= \frac{-t}{t^2+1} + \frac{2t}{(t^2+1)^2} - \arctan t + C \\
&= -\frac{1}{2}\sqrt{1-x^2} + \frac{\sqrt{1-x^2}}{t^2+1} - \arctan t + C \\
&= \frac{1}{2}\sqrt{1-x^2} \left(\frac{2}{t^2+1} - 1\right) - \arctan t + C \\
&= \frac{x\sqrt{1-x^2}}{2} - \frac{1}{2} \arccos x + C \\
&= \frac{x\sqrt{1-x^2}}{2} + \frac{1}{2} \arcsin x + K,
\end{aligned}$$

where for the very last equality we used the

fact that the derivatives of $\arcsin x$ and $\arccos x$ are negatives of one another.

Variant III. We show how to do this integral geometrically, provided already know the area of a sector of circle. Of course, here we assume we have already derived the formula for an area of a circle. We warn the reader that most methods for deriving the formula of a sector area rely on integrals, so it is possible we are making a circular reasoning argument. Since we already did the integral purely algebraically in the preceding solution variants, we can safely ignore the danger of the aforementioned circular reasoning argument. In other words, the present solution Variant is a geometric interpretation of the problem which relies on the formula for sector area of a circle (which we assumed proved elsewhere, possibly using similar integration techniques to the ones presented in Variant I and II).

By the Fundamental Theorem of Calculus, the indefinite integral measures up to a constant the area locked under the graph of $\sqrt{1-x^2}$. This graph is a part of a circle. Therefore, up to a constant, $\int \sqrt{1-t^2} dt$ equals $\int_0^x \sqrt{1-t^2} dt$. In turn $\int_0^x \sqrt{1-t^2} dt$ is given by the area highlighted in the picture below.

$$\begin{aligned}
\text{Area}(A) &= \frac{\text{length}(\widehat{PQ})}{2\pi} \pi = \frac{\text{length}(\widehat{PQ})}{2} = \frac{\arcsin x}{2} \\
\text{Area}(B) &= \text{Area}(\triangle OPQ) = \frac{x\sqrt{1-x^2}}{2} \\
\int_0^x \sqrt{1-t^2} dt &= \text{Area}(A) + \text{Area}(B) \\
&= \frac{\arcsin x}{2} + \frac{x\sqrt{1-x^2}}{2} \\
&\Rightarrow \\
\int \sqrt{1-x^2} dx &= \frac{\arcsin x}{2} + \frac{x\sqrt{1-x^2}}{2} + C \quad .
\end{aligned}$$

Solution. 24.e In this problem solution we use the standard Euler substitution $x = \cos(2 \arctan t)$. We recall from (??) that

$$\begin{aligned}
x &= \cos(2 \arctan t) = \frac{1-t^2}{1+t^2} \\
\arccos(x) &= 2 \arctan t \\
dx &= -\frac{4t}{(1+t^2)^2} dt \\
\sqrt{1-x^2} &= \sin(2 \arctan t) = \frac{2t}{1+t^2} \\
t &= \frac{\sqrt{1-x^2}}{x+1} \quad .
\end{aligned}$$

$$\begin{aligned}
\int \frac{\sqrt{1-x^2}}{1+x} dx &= \int t \left(-\frac{4t}{(1+t^2)^2} \right) dt & \left| \begin{array}{l} \text{Set } x = \frac{1-t^2}{1+t^2} \\ \text{Use f-las above} \end{array} \right. \\
&= -4 \int \frac{t^2}{(1+t^2)^2} dt \\
&= -4 \int \frac{1+t^2-1}{(1+t^2)^2} dt \\
&= -4 \int \left(\frac{1}{1+t^2} - \frac{1}{(1+t^2)^2} \right) dt \\
&= -4 \left(\arctan t - \frac{1}{2} \left(\arctan t + \frac{t}{1+t^2} \right) \right) + C \\
&= -2 \left(\arctan t - \frac{t}{1+t^2} \right) + C \\
&= -2 \left(\arctan \left(\frac{\sqrt{1-x^2}}{1+x} \right) - \frac{1}{2} \sqrt{1-x^2} \right) + C \\
&= -2 \arctan t + \sqrt{1-x^2} + C & \left| \begin{array}{l} \text{Use f-las above} \end{array} \right. \\
&= -\arccos x + \sqrt{1-x^2} + C \\
&= \arcsin x + \sqrt{1-x^2} + K \quad .
\end{aligned}$$

We have included the last equality to remind the student that derivatives of $\arcsin(x)$ and $\arccos x$ are negatives of one another.

25. Integrate

- (a) $\int \sqrt{x^2-1} dx$
- (b) $\int \sqrt{x^2-2} dx$
- (c) $\int \sqrt{2x^2+x-1} dx$
- (d) $\int \sqrt{x^2+x-1} dx$

26. (a) Express x , dx and $\sqrt{x^2+1}$ via θ and $d\theta$ for the trigonometric substitution $x = \cot \theta$, $\theta \in (0, \pi)$.

(b) Express x , dx and $\sqrt{x^2 + 1}$ via t and dt for the Euler substitution $x = \cot(2 \arctan t)$, $t > 0$. Express t via x .

Solution. 26.a The trigonometric substitution $x = \cot \theta$ is given by

$$\begin{aligned}\sqrt{x^2 + 1} &= \sqrt{\cot^2 \theta + 1} \\ &= \sqrt{\frac{\cos^2 \theta}{\sin^2 \theta} + 1} \\ &= \sqrt{\frac{\cos^2 \theta + \sin^2 \theta}{\sin^2 \theta}} \\ &= \sqrt{\frac{1}{\sin^2 \theta}} \quad \left| \begin{array}{l} \text{when } \theta \in (0, \pi) \text{ we have} \\ \sin \theta \geq 0 \text{ and so } \sqrt{\sin^2 \theta} = \sin \theta \end{array} \right. \\ &= \frac{1}{\sin \theta} = \csc \theta .\end{aligned}$$

The differential dx can be expressed via $d\theta$ from $x = \cot \theta$. The substitution $x = \cot \theta$ can be now summarized as:

$$\begin{aligned}x &= \cot \theta \\ \sqrt{x^2 + 1} &= \frac{1}{\sin \theta} = \csc \theta \\ dx &= -\frac{d\theta}{\sin^2 \theta} = -\csc^2 \theta d\theta \\ \theta &= \operatorname{arccot} x .\end{aligned}$$

Solution. 26.b We recall that the substitution $\theta = 2 \arctan t$ transforms a trigonometric integral into an integral of a rational function. We now apply the substitution $\theta = 2 \arctan t$ after the substitution $x = \cot \theta$:

$$\begin{aligned}x &= \cot \theta \\ &= \cot(2 \arctan t) \\ &= \frac{1 - \tan^2(\arctan t)}{2 \tan(\arctan t)} \\ &= \frac{1 - t^2}{2t} \\ &= \frac{1}{2} \left(\frac{1}{t} - t \right) .\end{aligned} \quad \left| \begin{array}{l} \text{use } \theta = 2 \arctan t \\ \text{use (??): } \cot 2z = \frac{\cos(2z)}{\sin(2z)} = \frac{1 - \tan^2 z}{2 \tan z} \end{array} \right.$$

We can furthermore compute

$$\begin{aligned}\sqrt{x^2 + 1} &= \sqrt{\frac{1}{4} \left(\frac{1}{t} - t \right)^2 + 1} \\ &= \frac{1}{2} \sqrt{\left(\frac{1}{t} + t \right)^2} \quad \left| \sqrt{\left(\frac{1}{t} + t \right)^2} = \frac{1}{t} + t \text{ because } t > 0 \right. \\ &= \frac{1}{2} \left(\frac{1}{t} + t \right) .\end{aligned} \tag{3}$$

The differential dx can via dx as follows.

$$dx = d \left(\frac{1}{2} \left(\frac{1}{t} - t \right) \right) = -\frac{1}{2} \left(\frac{1}{t^2} - 1 \right) dt .$$

Finally, we can subtract $x = \frac{1}{2} \left(\frac{1}{t} - t \right)$ from $\sqrt{x^2 + 1} = \frac{1}{2} \left(\frac{1}{t} + t \right)$ to get that

$$t = \sqrt{x^2 + 1} - x .$$

The Euler substitution $x = \cot \theta = \cot(\arctan 2t)$ can be now summarized as:

$$\begin{aligned}x &= \frac{1}{2} \left(\frac{1}{t} - t \right) \\ \sqrt{x^2 + 1} &= \frac{1}{2} \left(\frac{1}{t} + t \right) \\ dx &= -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\ t &= \sqrt{x^2 + 1} - x .\end{aligned} \tag{4}$$

27. Let the variables x and t be related via $\sqrt{x^2 + 1} = x + t$.

- (a) Express x via t .
- (b) Express $\sqrt{x^2 + 1}$ via t alone.
- (c) Express dx via t and dt .

Solution. 27.a.

$$\begin{aligned} \sqrt{x^2 + 1} &= x + t & \Big| \text{ square both sides} \\ x^2 + 1 &= x^2 + 2xt + t^2 \\ -2xt &= t^2 - 1 \\ x &= \frac{1}{2} \left(\frac{1}{t} - t \right) . \end{aligned}$$

Solution. 27.b.

Use Problem 27.a to get:

$$\sqrt{x^2 + 1} = x + t = \frac{1}{2} \left(\frac{1}{t} - t \right) + t = \frac{1}{2} \left(\frac{1}{t} + t \right) .$$

28. (a) Express x , dx and $\sqrt{x^2 + 1}$ via θ and $d\theta$ for the trigonometric substitution $x = \tan \theta$, $\theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.
- (b) Express x , dx and $\sqrt{x^2 + 1}$ via t and dt for the Euler substitution $x = \tan(2 \arctan t)$, $t \in (-1, 1)$. Express t via x .

Solution. 28.a The trigonometric substitution $x = \tan \theta$ is given by

$$\begin{aligned} \sqrt{x^2 + 1} &= \sqrt{\tan^2 \theta + 1} \\ &= \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta} + 1} \\ &= \sqrt{\frac{\sin^2 \theta + \cos^2 \theta}{\cos^2 \theta}} \\ &= \sqrt{\frac{1}{\cos^2 \theta}} & \Bigg| \begin{array}{l} \text{when } \theta \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \text{ we have} \\ \cos \theta > 0 \text{ and so } \sqrt{\cos^2 \theta} = \cos \theta \end{array} \\ &= \frac{1}{\cos \theta} = \sec \theta . \end{aligned}$$

The differential dx can be expressed via $d\theta$ from $x = \tan \theta$. The substitution $x = \tan \theta$ can be now summarized as:

$$\begin{aligned} x &= \tan \theta \\ \sqrt{x^2 + 1} &= \frac{1}{\cos \theta} = \sec \theta \\ dx &= \frac{d\theta}{\cos^2 \theta} = \sec^2 \theta d\theta \\ \theta &= \arctan x . \end{aligned}$$

Solution. 28.b We recall that the substitution $\theta = 2 \arctan t$ transforms a trigonometric integral into an integral of a rational function. We now apply the substitution $\theta = 2 \arctan t$ after the substitution $x = \tan \theta$:

$$\begin{aligned} x &= \tan \theta & \Bigg| \begin{array}{l} \text{use } \theta = 2 \arctan t \\ \text{use (??): } \tan 2z = \frac{\sin(2z)}{\cos(2z)} = \frac{2 \tan z}{1 - \tan^2 z} \end{array} \\ &= \tan(2 \arctan t) \\ &= \frac{2 \tan(\arctan t)}{1 - \tan^2(\arctan t)} \\ &= \frac{2t}{1 - t^2} . \end{aligned}$$

We can furthermore compute

$$\begin{aligned}
 \sqrt{x^2 + 1} &= \sqrt{\left(\frac{2t}{1-t^2}\right)^2 + 1} \\
 &= \sqrt{\frac{4t^2 + (1-t^2)^2}{(1-t^2)^2}} \\
 &= \frac{\sqrt{(1+t^2)^2}}{(1-t^2)} \quad \left| \sqrt{(1-t^2)^2} = 1-t^2 \text{ because } |t| < 1 \right. \\
 &= \frac{1+t^2}{1-t^2} \\
 &= \frac{1-t^2}{2-(1-t^2)} \\
 &= \frac{1-t^2}{1-t^2} \\
 &= -1 + \frac{2}{1-t^2} .
 \end{aligned} \tag{5}$$

From $\sqrt{x^2 + 1} = -1 + \frac{2}{1-t^2}$ and $x = \frac{2t}{1-t^2}$ we can express t via x :

$$\begin{aligned}
 \sqrt{x^2 + 1} &= -1 + \frac{2}{1-t^2} \\
 &= -1 + \frac{1}{t} \left(\frac{2t}{1-t^2} \right) \quad \left| \text{use } x = \frac{2t}{1-t^2} \right. \\
 &= -1 + \frac{x}{t} \\
 1 + \sqrt{x^2 + 1} &= \frac{x}{t} \\
 t &= \frac{x}{1 + \sqrt{x^2 + 1}} \\
 &= \frac{x}{1 + \sqrt{x^2 + 1}} \left(\frac{1 - \sqrt{x^2 + 1}}{1 - \sqrt{x^2 + 1}} \right) \\
 &= \frac{x(1 - \sqrt{x^2 + 1})}{1 - x^2 - 1} \\
 &= \frac{\sqrt{x^2 + 1} - 1}{x} .
 \end{aligned}$$

The differential dx can be expressed via dt from $x = \frac{2t}{1-t^2}$. The Euler substitution $x = \tan \theta = \tan(2 \arctan t)$ can now be summarized as follows.

$$\begin{aligned}
 x &= \frac{2t}{1-t^2} \\
 \sqrt{x^2 + 1} &= -1 + \frac{2}{1-t^2} \\
 dx &= \frac{2(1+t^2)}{(1-t^2)^2} dt \\
 t &= \frac{\sqrt{x^2 + 1} - 1}{x} .
 \end{aligned} \tag{6}$$

29. Let the variables x and t be related via $\sqrt{x^2 + 1} = \frac{x}{t} - 1$.

- (a) Express x via t .
- (b) Express $\sqrt{x^2 + 1}$ via t alone.
- (c) Express dx via t and dt .

30. (a) Express x , dx and $\sqrt{1-x^2}$ via θ and $d\theta$ for the trigonometric substitution $x = \cos \theta$, $\theta \in [0, \pi]$.
 (b) Express x , dx and $\sqrt{1-x^2}$ via t and dt for the Euler substitution $x = \cos(2 \arctan t)$, $t \geq 0$. Express t via x .

Solution. 30.a The trigonometric substitution $x = \cos \theta$ is given by

$$\begin{aligned}
 \sqrt{-x^2 + 1} &= \sqrt{1 - \cos^2 \theta} \\
 &= \sqrt{\sin^2 \theta} \quad \left| \begin{array}{l} \text{when } \theta \in [0, \pi] \text{ we have} \\ \sin \theta \geq 0 \text{ and so } \sqrt{\sin^2 \theta} = \sin \theta \end{array} \right. \\
 &= \sin \theta .
 \end{aligned}$$

The differential dx can be expressed via $d\theta$ from $x = \cos \theta$. The substitution $x = \cos \theta$ can be now summarized as:

$$\begin{aligned} x &= \cos \theta \\ \sqrt{-x^2 + 1} &= \sin \theta \\ dx &= -\sin \theta d\theta \\ \theta &= \arccos x \quad . \end{aligned}$$

Solution. 30.b We recall that the substitution $\theta = 2 \arctan t$ transforms a trigonometric integral into an integral of a rational function. We now apply the substitution $2 \arctan t$ after the substitution $x = \cos \theta$:

$$\begin{aligned} x &= \cos \theta \\ &= \cos(2 \arctan t) & \left| \begin{array}{l} \text{use } \theta = 2 \arctan t \\ \text{use(??): } \cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z} \end{array} \right. \\ &= \frac{1 - \tan^2(\arctan t)}{1 + \tan^2(\arctan t)} \\ &= \frac{1 - t^2}{1 + t^2} \quad . \end{aligned}$$

We can furthermore compute

$$\begin{aligned} \sqrt{-x^2 + 1} &= \sqrt{1 - \left(\frac{1 - t^2}{1 + t^2}\right)^2} \\ &= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 + t^2)^2}} \\ &= \sqrt{\frac{4t^2}{(1 + t^2)^2}} & \left| \begin{array}{l} \sqrt{4t^2} = 2t \text{ because } t \geq 0 \end{array} \right. \\ &= \frac{2t}{1 + t^2} \quad . \end{aligned} \tag{7}$$

The differential dx can be computed from $x = \frac{1-t^2}{1+t^2}$. Finally, we can express t via x with a little algebra:

$$\begin{aligned} x &= \frac{1 - t^2}{1 + t^2} \\ (1 + t^2)x &= 1 - t^2 \\ t^2(x + 1) &= 1 - x \\ t^2 &= \frac{1 - x}{1 + x} \\ t &= \sqrt{\frac{1 - x}{1 + x}} & \left| \begin{array}{l} \text{here we use } t > 0 \end{array} \right. \\ t &= \frac{\sqrt{1 - x} \sqrt{1 + x}}{\sqrt{1 + x} \sqrt{1 + x}} \\ t &= \frac{\sqrt{-x^2 + 1}}{x + 1} \quad . \end{aligned}$$

The Euler substitution $x = \cos(2 \arctan t)$ can be now summarized as:

$$\begin{aligned} x &= \frac{1 - t^2}{1 + t^2} \\ \sqrt{-x^2 + 1} &= \frac{2t}{1 + t^2} \\ dx &= -\frac{4t}{(t^2 + 1)^2} dt \\ t &= \frac{\sqrt{-x^2 + 1}}{x + 1} \quad . \end{aligned} \tag{8}$$

31. Let the variables x and t be related via $\sqrt{-x^2 + 1} = (1 - x)t$.

(a) Express x via t .

(b) Express $\sqrt{-x^2 + 1}$ via t alone.

(c) Express dx via t and dt .

Solution. 31.a.

$$\begin{array}{rcl}
 \sqrt{-x^2 + 1} & = & (1-x)t \\
 (1-x)(1+x) & = & (1-x)^2 t^2 \\
 1+x & = & (1-x)t^2 \\
 x(1+t^2) & = & t^2 - 1 \\
 x & = & \frac{t^2 - 1}{t^2 + 1} = 1 - \frac{2}{t^2 + 1} \quad .
 \end{array}
 \quad \left| \begin{array}{l} \text{square both sides} \\ \text{divide by } (1-x) \end{array} \right.$$

Solution. 31.b.

Use Problem 31.a to get

$$\sqrt{-x^2 + 1} = (1-x)t = \left(1 - \left(1 - \frac{2t}{t^2 + 1}\right)\right)t = \frac{2t}{t^2 + 1} \quad .$$

32. (a) Express x , dx and $\sqrt{1-x^2}$ via θ and $d\theta$ for the trigonometric substitution $x = \sin \theta$, $\theta \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

(b) Express x , dx and $\sqrt{1-x^2}$ via t and dt for the Euler substitution $x = \sin(2 \arctan t)$, $t \in [-1, 1]$. Express t via x .

Solution. 32.a The trigonometric substitution $x = \sin \theta$ is given by

$$\begin{array}{rcl}
 \sqrt{-x^2 + 1} & = & \sqrt{1 - \sin^2 \theta} \\
 & = & \sqrt{\cos^2 \theta} \\
 & = & \cos \theta \quad .
 \end{array}
 \quad \left| \begin{array}{l} \text{when } \theta \in [-\frac{\pi}{2}, \frac{\pi}{2}] \text{ we have} \\ \cos \theta \geq 0 \text{ and so } \sqrt{\cos^2 \theta} = \cos \theta \end{array} \right.$$

The differential dx can be expressed via $d\theta$ from $x = \sin \theta$. The substitution $x = \sin \theta$ can be now summarized as:

$$\begin{array}{rcl}
 x & = & \sin \theta \\
 \sqrt{-x^2 + 1} & = & \sin \theta \\
 dx & = & \cos \theta d\theta \\
 \theta & = & \arcsin x \quad .
 \end{array}$$

Solution. 32.b We recall that the substitution $\theta = 2 \arctan t$ transforms a trigonometric integral into an integral of a rational function. We now apply the substitution $2 \arctan t$ after the substitution $x = \sin \theta$:

$$\begin{array}{rcl}
 x & = & \sin \theta \\
 & = & \sin(2 \arctan t) \\
 & = & \frac{2 \tan(\arctan t)}{1 + \tan^2(\arctan t)} \\
 & = & \frac{2t}{1 + t^2} \quad .
 \end{array}
 \quad \left| \begin{array}{l} \text{use } \theta = 2 \arctan t \\ \text{use(??) : } \sin(2z) = \frac{2 \tan z}{1 + \tan^2 z} \end{array} \right.$$

We can furthermore compute

$$\begin{array}{rcl}
 \sqrt{-x^2 + 1} & = & \sqrt{1 - \left(\frac{2t}{1+t^2}\right)^2} \\
 & = & \sqrt{\frac{(1+t^2)^2 - 4t^2}{(1+t^2)^2}} \\
 & = & \sqrt{\frac{(1-t^2)^2}{(1+t^2)^2}} \\
 & = & \frac{1-t^2}{1+t^2} \\
 & = & \frac{2 - (1+t^2)}{1+t^2} \\
 & = & -1 + \frac{2}{1+t^2} \quad .
 \end{array}
 \quad \left| \begin{array}{l} \sqrt{(1-t^2)^2} = 1-t^2 \text{ because } |t| \leq 1 \end{array} \right. \quad (9)$$

The differential dx can be computed from $x = \frac{2t}{1+t^2}$. Finally, we can express t via x with a little algebra:

$$\begin{aligned}
 \sqrt{-x^2+1} &= -1 + \frac{2}{1+t^2} \\
 &= -1 + \frac{1}{t} \left(\frac{2t}{1+t^2} \right) && \left| \begin{array}{l} \text{use } x = \frac{2t}{1+t^2} \\ +1 \text{ to both sides} \end{array} \right. \\
 &= -1 + \frac{x}{t} \\
 \frac{x}{t} &= 1 + \sqrt{-x^2+1} \\
 t &= \frac{x}{1 + \sqrt{-x^2+1}} \\
 &= \frac{x}{(1 + \sqrt{-x^2+1})(1 - \sqrt{-x^2+1})} \\
 &= \frac{x}{1 - (-x^2+1)} \\
 &= \frac{x}{x} .
 \end{aligned}$$

The Euler substitution $x = \sin(2 \arctan t)$ can be now summarized as:

$$\begin{aligned}
 x &= \frac{2t}{1+t^2} \\
 \sqrt{-x^2+1} &= -1 + \frac{2}{1+t^2} \\
 dx &= 2 \left(\frac{1-t^2}{(1+t^2)^2} \right) dt \\
 t &= \frac{1 - \sqrt{-x^2+1}}{x} .
 \end{aligned}$$

33. Let the variables x and t be related via $\sqrt{-x^2+1} = 1 - xt$.

- (a) Express x via t .
- (b) Express $\sqrt{-x^2+1}$ via t alone.
- (c) Express dx via t and dt .

34. (a) Express x , dx and $\sqrt{x^2-1}$ via θ and $d\theta$ for the trigonometric substitution $x = \sec \theta$, $\theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2})$.
 (b) Express x , dx and $\sqrt{1-x^2}$ via t and dt for the Euler substitution $x = \sec(2 \arctan t)$, $t \in (-\infty, -1) \cup [1, 0)$. Express t via x .

Solution. 34.a The trigonometric substitution $x = \sec \theta$ is given by

$$\begin{aligned}
 \sqrt{x^2-1} &= \sqrt{\sec^2 \theta - 1} = \sqrt{\frac{1}{\cos^2 \theta} - 1} \\
 &= \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}} = \sqrt{\tan^2 \theta} && \left| \begin{array}{l} \text{when } \theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2}) \text{ we have} \\ \tan \theta \geq 0 \text{ and so } \sqrt{\tan^2 \theta} = \tan \theta \end{array} \right. \\
 &= \tan \theta .
 \end{aligned}$$

The differential dx can be expressed via $d\theta$ from $x = \sec \theta$. The substitution $x = \sec \theta$ can be now summarized as:

$$\begin{aligned}
 x &= \sec \theta = \frac{1}{\cos \theta} \\
 \sqrt{x^2-1} &= \tan \theta \\
 dx &= \frac{\sin \theta}{\cos^2 \theta} d\theta = \sec \theta \tan \theta d\theta \\
 \theta &= \operatorname{arcsec} x .
 \end{aligned}$$

Solution. 34.b We recall that the substitution $\theta = 2 \arctan t$ transforms a trigonometric integral into an integral of a rational function. We now apply the substitution $2 \arctan t$ after the substitution $x = \sec \theta$:

$$\begin{aligned}
x &= \sec \theta = \frac{1}{\cos \theta} & \left| \begin{array}{l} \text{use } \theta = 2 \arctan t \\ \text{use (??): } \cos(2z) = \frac{1 - \tan^2 z}{1 + \tan^2 z} \end{array} \right. \\
&= \frac{1}{\cos(2 \arctan t)} \\
&= \frac{1 + \tan^2(\arctan t)}{1 - \tan^2(\arctan t)} \\
&= \frac{1 + t^2}{1 - t^2} \\
&= -1 + \frac{2}{1 - t^2} .
\end{aligned}$$

We can furthermore compute

$$\begin{aligned}
\sqrt{x^2 - 1} &= \sqrt{\left(\frac{1 + t^2}{1 - t^2}\right)^2 - 1} \\
&= \sqrt{\frac{(1 + t^2)^2 - (1 - t^2)^2}{(1 - t^2)^2}} \\
&= \sqrt{\frac{4t^2}{(1 - t^2)^2}} & \left| \begin{array}{l} t \text{ and } 1 - t^2 \text{ have the same} \\ \text{sign for } t \in (-\infty, -1) \cup [0, 1) \end{array} \right. \\
&= \frac{2t}{1 - t^2} .
\end{aligned} \tag{10}$$

The differential dx can be computed from $x = \frac{1+t^2}{1-t^2}$. Finally, we can express t via x with a little algebra:

$$\begin{aligned}
x &= \frac{1 + t^2}{1 - t^2} \\
(1 - t^2)x &= 1 + t^2 \\
(1 + x)t^2 &= x - 1 \\
t^2 &= \frac{x - 1}{x + 1} \\
t &= \begin{cases} \sqrt{\frac{x-1}{x+1}} & x > 1 \\ -\sqrt{\frac{x-1}{x+1}} & x < -1 \end{cases} & \left| \begin{array}{l} \text{because when } x < -1, \\ \text{we have } t \in (-\infty, -1] \end{array} \right. \\
t &= \begin{cases} \frac{\sqrt{x^2-1}}{x+1} & x > 1 \\ -\frac{\sqrt{x^2-1}}{x+1} & x < -1 \end{cases} .
\end{aligned}$$

The Euler substitution $x = \sec(2 \arctan t)$ can be now summarized as:

$$\begin{aligned}
x &= \frac{1 + t^2}{1 - t^2} \\
\sqrt{x^2 - 1} &= \frac{2t}{1 - t^2} \\
dx &= \frac{4t}{(1 - t^2)^2} dt \\
t &= \pm \frac{\sqrt{x^2 - 1}}{x + 1} .
\end{aligned}$$

35. Let the variables x and t be related via $\sqrt{x^2 - 1} = (x + 1)t$.

- (a) Express x via t .
- (b) Express $\sqrt{x^2 - 1}$ via t alone.
- (c) Express dx via t and dt .

Solution. 35.a.

$$\begin{array}{rcl}
\sqrt{x^2 - 1} & = & (x + 1)t \\
(x - 1)(x + 1) & = & (x + 1)^2 t^2 \\
x - 1 & = & (x + 1)t^2 \\
x(1 - t^2) & = & 1 + t^2 \\
x & = & \frac{1 + t^2}{1 - t^2} = -1 + \frac{2}{1 - t^2}
\end{array} \quad \left| \begin{array}{l} \text{square both sides} \\ \text{divide by } (x + 1) \end{array} \right.$$

Solution. 35.b.

We use Problem 35.a to get

$$\sqrt{x^2 - 1} = (x + 1)t = \left(-1 + \frac{2}{1 - t^2} + 1\right)t = \frac{2t}{1 - t^2}$$

36. (a) Express x , dx and $\sqrt{1 - x^2}$ via θ and $d\theta$ for the trigonometric substitution $x = \csc \theta$, $\theta \in [0, \frac{\pi}{2}] \cup [\pi, \frac{3\pi}{2})$.
(b) Express x , dx and $\sqrt{1 - x^2}$ via t and dt for the Euler substitution $x = \csc(2 \arctan t)$, $t \in (-\infty, -1) \cup [0, 1)$. Express t via x .

Solution. 36.a The trigonometric substitution $x = \csc \theta$ is given by

$$\begin{array}{rcl}
\sqrt{x^2 - 1} & = & \sqrt{\frac{1}{\sin^2 \theta} - 1} \\
& = & \sqrt{\frac{\cos^2 \theta}{\sin^2 \theta}} = \sqrt{\cot^2 \theta} \\
& = & \cot \theta
\end{array} \quad \left| \begin{array}{l} \text{when } \theta \in [0, \frac{\pi}{2}) \cup [\pi, \frac{3\pi}{2}) \text{ we have} \\ \cot \theta \geq 0 \text{ and so } \sqrt{\cot^2 \theta} = \cot \theta \end{array} \right.$$

The differential dx can be expressed via $d\theta$ from $x = \csc \theta$. The substitution $x = \csc \theta$ can be now summarized as:

$$\begin{array}{rcl}
x & = & \csc \theta \\
\sqrt{x^2 - 1} & = & \cot \theta \\
dx & = & -\frac{\cos \theta}{\sin^2 \theta} d\theta = -\csc \theta \cot \theta d\theta \\
\theta & = & \csc^{-1} x
\end{array}$$

Solution. 36.b We recall that the substitution $\theta = 2 \arctan t$ transforms a trigonometric integral into an integral of a rational function. We now apply the substitution $2 \arctan t$ after the substitution $x = \csc \theta$:

$$\begin{array}{rcl}
x & = & \csc \theta = \frac{1}{\sin \theta} \\
& = & \frac{1}{\sin(2 \arctan t)} \\
& = & \frac{1}{\frac{2 \tan(\arctan t)}{1 + \tan^2(\arctan t)}} \\
& = & \frac{1 + t^2}{2t} \\
& = & \frac{1}{2} \left(\frac{1}{t} + t \right)
\end{array} \quad \left| \begin{array}{l} \text{use } \theta = 2 \arctan t \\ \text{use(??): } \sin(2z) = \frac{2 \tan z}{1 + \tan^2 z} \end{array} \right.$$

We can furthermore compute

$$\begin{array}{rcl}
\sqrt{x^2 - 1} & = & \sqrt{\left(\frac{1 + t^2}{2t}\right)^2 - 1} \\
& = & \sqrt{\frac{(1 + t^2)^2 - 4t^2}{4t^2}} \\
& = & \sqrt{\frac{(1 - t^2)^2}{4t^2}} \\
& = & \frac{1 - t^2}{2t} \\
& = & \frac{1}{2} \left(\frac{1}{t} - t \right)
\end{array} \quad \left| \begin{array}{l} \frac{1 - t^2}{2t} > 0 \text{ when } t \in (-\infty, -1) \cup [0, 1) \end{array} \right. \quad (11)$$

The differential dx can be computed from $x = \frac{1}{2} \left(\frac{1}{t} - t \right)$. Finally, we can express t via x with a little algebra:

$$\begin{aligned} \sqrt{x^2 - 1} &= \frac{1 - t^2}{2t} \\ \sqrt{x^2 - 1} &= \frac{2 - (1 + t^2)}{2t} & \left| \text{use } x = \frac{1 + t^2}{2t} \right. \\ \sqrt{x^2 - 1} &= \frac{1}{t} - x \\ \frac{1}{t} &= \sqrt{x^2 - 1} + x \\ t &= \frac{1}{\sqrt{x^2 - 1} + x} = \frac{1}{(\sqrt{x^2 - 1} + x)(-\sqrt{x^2 - 1} + x)} \\ t &= x - \sqrt{x^2 - 1} \end{aligned}$$

The Euler substitution $x = \cos(2 \arctan t)$ can be now summarized as:

$$\begin{aligned} x &= \frac{1}{2} \left(\frac{1}{t} + t \right) \\ \sqrt{-x^2 + 1} &= \frac{1}{2} \left(\frac{1}{t} - t \right) \\ dx &= -\frac{1}{2} \left(\frac{1}{t^2} + 1 \right) dt \\ t &= x - \sqrt{x^2 - 1} \end{aligned}$$

37. Let the variables x and t be related via $\sqrt{x^2 - 1} = \frac{1}{t} - x$.

- Express x via t .
- Express $\sqrt{x^2 - 1}$ via t alone.
- Express dx via t and dt .

38. Compute the limits. The answer key has not been fully proofread, use with caution.

(a) $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

ANSWER: 1

(b) $\lim_{x \rightarrow 0} \frac{x}{\ln(1+x)}$.

ANSWER: 1

(c) $\lim_{x \rightarrow 0} \frac{x^2}{x - \ln(1+x)}$.

ANSWER: 2

(d) $\lim_{x \rightarrow 0} \frac{x^2}{\sin x \ln(1+x)}$.

ANSWER: 1

(e) $\lim_{x \rightarrow 0} \frac{\sin^2 x}{(\ln(1+x))^2}$.

ANSWER: 1

(f) $\lim_{x \rightarrow 0} \frac{\cos x - 1}{\sin x \ln(1+x)}$.

ANSWER: $\frac{2}{1}$

(g) $\lim_{x \rightarrow 0} \frac{\arctan x - x}{x^3}$.

ANSWER: $-\frac{3}{1}$

(h) $\lim_{x \rightarrow 0} \frac{\arcsin x - x}{x^3}$.

ANSWER: $\frac{0}{1}$

(i) $\lim_{x \rightarrow 1} \frac{x}{x-1} - \frac{1}{\ln x}$.

ANSWER: $\frac{2}{1}$

(j) $\lim_{x \rightarrow 0} \frac{\cos(nx) - \cos(mx)}{x^2}$.

ANSWER: $\frac{m^2 - n^2}{2}$

(k) $\lim_{x \rightarrow 0} \frac{\arcsin x - x - \frac{1}{6}x^3}{\sin^5 x}$.

ANSWER: $\frac{0}{3}$

(l) $\lim_{x \rightarrow 1} \frac{\sin(\pi x) \ln x}{\cos(\pi x) + 1}$.

ANSWER: $-\frac{\pi}{2}$

(m) $\lim_{x \rightarrow 0} \frac{\sin x - x}{\arcsin x - x}$.

ANSWER: -1

(n) $\lim_{x \rightarrow 0} \frac{\sin x - x}{\arctan x - x}$.

ANSWER: $\frac{2}{1}$

(o) $\lim_{x \rightarrow \infty} x \sin\left(\frac{2}{x}\right)$.

ANSWER: 2

Solution. 38l The limit is of the form “ $\frac{0}{0}$ ” so we are allowed to use L’Hospital’s rule.

$$\begin{aligned}
 \lim_{x \rightarrow 1} \frac{\sin(\pi x) \ln x}{\cos(\pi x) + 1} &= \lim_{x \rightarrow 1} \frac{(\sin(\pi x) \ln x)'}{(\cos(\pi x) + 1)'} \\
 &= \lim_{x \rightarrow 1} \frac{(\pi \cos(\pi x) \ln x + \sin(\pi x) \frac{1}{x})}{(-\pi \sin(\pi x))} \quad \left| \begin{array}{l} \text{type “}\frac{0}{0}\text{”, L’Hospital’s rule} \end{array} \right. \\
 &= \lim_{x \rightarrow 1} \frac{(\pi \cos(\pi x) \ln x + \sin(\pi x) \frac{1}{x})'}{(-\pi \sin(\pi x))'} \\
 &= \lim_{x \rightarrow 1} \frac{-\pi^2 \sin(\pi x) \ln(x) + 2\pi \cos(\pi x) x^{-1} - \sin(\pi x) x^{-2}}{(-\pi^2 \cos(\pi x))} \\
 &= \frac{-\pi^2 \sin(\pi) \ln(1) + 2\pi \cos(\pi) - \sin(\pi)}{(-\pi^2 \cos(\pi))} \\
 &= -\frac{2}{\pi} .
 \end{aligned}$$

Solution. 38n **Solution I.**

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\sin x - x}{\arctan x - x} &= \lim_{x \rightarrow 0} \frac{\cos x - 1}{\frac{1}{1+x^2} - 1} \quad \left| \begin{array}{l} \text{L’Hospital rule} \\ \text{L’Hospital rule again} \end{array} \right. \\
 &= \lim_{x \rightarrow 0} \frac{-\sin x}{\frac{-2x}{(1+x^2)^2}} \\
 &= \lim_{x \rightarrow 0} \frac{(1+x^2)^2 \sin x}{2x} \\
 &= \lim_{x \rightarrow 0} \frac{(1+x^2)^2}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x} \\
 &= \frac{1}{2} .
 \end{aligned}$$

Solution II.

$$\begin{aligned}
 \lim_{x \rightarrow 0} \frac{\sin x - x}{\arctan x - x} &= \lim_{x \rightarrow 0} \frac{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots\right) - x}{\left(x - \frac{x^3}{3} + \frac{x^5}{5} - \dots\right) - x} \quad \left| \begin{array}{l} \text{use the Maclaurin series of } \sin, \arctan \\ \text{The expressions in parenthesis} \\ \text{are continuous functions in } x \end{array} \right. \\
 &= \lim_{x \rightarrow 0} \frac{-\frac{x^3}{6} + x^5 \left(\frac{1}{5!} - \dots\right)}{-\frac{x^3}{3} + x^5 \left(\frac{1}{5} - \dots\right)} \\
 &= \lim_{x \rightarrow 0} \frac{-\frac{1}{6} + x^2 \left(\frac{1}{5!} - \dots\right)}{-\frac{1}{3} + x^2 \left(\frac{1}{5} - \dots\right)} \\
 &= \frac{-\frac{1}{6} + 0}{-\frac{1}{3} + 0} \\
 &= \frac{1}{2} .
 \end{aligned}$$

Solution. 38o.

$$\begin{aligned}
 \lim_{x \rightarrow \infty} x \sin\left(\frac{2}{x}\right) &= \lim_{x \rightarrow \infty} \frac{\sin\left(\frac{2}{x}\right)}{\frac{1}{x}} \quad \left| \begin{array}{l} \text{indeterminate form} \\ \text{Use L’Hospital’s rule} \end{array} \right. \\
 &= \lim_{x \rightarrow \infty} \frac{\cos\left(\frac{2}{x}\right) \left(-\frac{2}{x^2}\right)}{-\frac{1}{x^2}} \\
 &= \lim_{x \rightarrow \infty} 2 \cos\left(\frac{2}{x}\right) \\
 &= 2 .
 \end{aligned}$$

39. Compute the limit.

(a) $\lim_{x \rightarrow \infty} \left(\frac{x-2}{x}\right)^x .$

(b) $\lim_{x \rightarrow \infty} \left(\frac{x-2}{x}\right)^{2x}$

$$(c) \lim_{x \rightarrow \infty} \left(\frac{x}{x+3} \right)^{2x}$$

answer: 9

Solution. 39.a.

Variant I.

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(\frac{x-2}{x} \right)^x &= \lim_{x \rightarrow \infty} \left(1 - \frac{2}{x} \right)^x \quad \left| \text{ use } \lim_{x \rightarrow \infty} \left(1 + \frac{k}{x} \right)^x = e^k \right. \\ &= e^{-2} . \end{aligned}$$

Variant II.

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(\frac{x-2}{x} \right)^x &= \lim_{x \rightarrow \infty} e^{\ln\left(\left(\frac{x-2}{x}\right)^x\right)} \\ \lim_{x \rightarrow \infty} \ln\left(\left(\frac{x-2}{x}\right)^x\right) &= \lim_{x \rightarrow \infty} x(\ln(x-2) - \ln(x)) \\ &= \lim_{x \rightarrow \infty} \frac{\ln(x-2) - \ln(x)}{\frac{1}{x}} \quad \left| \text{ L'Hospital rule} \right. \\ &= \lim_{x \rightarrow \infty} \frac{\frac{1}{x-2} - \frac{1}{x}}{-\frac{1}{x^2}} \\ &= \lim_{x \rightarrow \infty} \frac{-2x^2}{x^2 - 2x} = -2 \quad \left| \text{ Therefore} \right. \\ \lim_{x \rightarrow \infty} \left(\frac{x-2}{x} \right)^x &= \lim_{x \rightarrow \infty} e^{\ln\left(\left(\frac{x-2}{x}\right)^x\right)} \\ &= e^{\lim_{x \rightarrow \infty} \ln\left(\left(\frac{x-2}{x}\right)^x\right)} \\ &= e^{-2} . \end{aligned}$$

40. Find the limit.

$$(a) \lim_{x \rightarrow \infty} \left(1 - \frac{2}{x} \right)^x .$$

answer: e^{-2}

$$(c) \lim_{x \rightarrow \infty} \left(\frac{x}{x-5} \right)^x .$$

answer: e^5

$$(b) \lim_{x \rightarrow 0} (1-x)^{\frac{1}{x}} .$$

answer: e^{-1}

$$(d) \lim_{x \rightarrow \infty} \left(\frac{x}{x-2} \right)^{3x+2} .$$

answer: e^6

41. Determine whether the integral is convergent or divergent. Motivate your answer.

$$(a) \int_2^{\infty} \frac{1}{(x-1)^{\frac{3}{2}}} dx .$$

answer: convergent

$$(g) \int_{-\infty}^0 \frac{1}{(2-3x)^{1.00000001}} dx .$$

answer: convergent

$$(b) \int_{-1}^1 \frac{1}{\sqrt[5]{1+x}} dx .$$

answer: convergent

$$(h) \int_{-2}^{\frac{1}{2}} \frac{1}{2x-1} dx .$$

answer: divergent

$$(c) \int_1^{\infty} \frac{1}{\sqrt[5]{1+x}} dx .$$

answer: divergent

$$(i) \int_{-1}^{\infty} e^{-3x} dx .$$

answer: convergent, equals $\frac{3}{e}$

$$(d) \int_{-1}^{\infty} \frac{1}{\sqrt[5]{1+x}} dx .$$

answer: divergent

$$(j) \int_{-\infty}^5 2^x dx .$$

answer: convergent

$$(e) \int_{-\infty}^0 \frac{1}{2-3x} dx .$$

answer: divergent

$$(k) \int_{-\infty}^{\infty} x^3 dx .$$

answer: divergent

$$(f) \int_{-\infty}^0 \frac{1}{(2-3x)^2} dx .$$

answer: convergent

$$(l) \int_{-\infty}^{\infty} x e^{-x^2} dx .$$

(m) $\int_0^{\infty} \sqrt{x} e^{-\sqrt{x}} dx.$	answer: divergent	(t) $\int_{100}^{\infty} \frac{1}{x \ln x} dx.$	answer: convergent, equals 0
(n) $\int_0^{\infty} \sin^2 x dx.$	answer: convergent	(u) $\int_{100}^{\infty} \frac{1}{x(\ln x)^2} dx.$	answer: convergent, equals 4
(o) $\int_0^5 \frac{1}{x^2 + x - 2} dx.$	answer: divergent	(v) $\int_0^1 \ln x dx.$	answer: convergent
(p) $\int_0^{\infty} \frac{1}{x^2 + x + 1} dx.$	answer: divergent	(w) $\int_0^1 \frac{\ln x}{\sqrt{x}} dx.$	answer: convergent
(q) $\int_2^{\infty} \frac{1}{x^2 - x - 1} dx.$	answer: convergent	(x) $\int_0^2 x^3 \ln x dx.$	answer: convergent, equals $-1 + 4 \ln 2$
(r) $\int_0^{\infty} \frac{1}{x^2 - x - 1} dx.$	answer: convergent	(y) $\int_0^1 \frac{e^{\frac{1}{x}}}{x^2} dx.$	answer: divergent
(s) $\int_{-\infty}^{\infty} \frac{x^2}{x^4 + 2} dx.$	answer: convergent	(z) $\int_{-1}^0 \frac{e^{\frac{1}{x}}}{x^2} dx.$	answer: convergent

Solution. 41.m It is possible to show that this integral is convergent by using the comparison theorem. However, we shall use direct integration instead. First, we solve the indefinite integral:

$$\begin{aligned}
 \int \sqrt{x} e^{-\sqrt{x}} dx &= \int \sqrt{x} e^{-\sqrt{x}} \frac{2\sqrt{x} dx}{2\sqrt{x}} && \left| \begin{array}{l} \text{use } d\sqrt{x} = \frac{dx}{2\sqrt{x}} \\ \text{Set } \sqrt{x} = u \end{array} \right. \\
 &= \int \sqrt{x} e^{-\sqrt{x}} (2\sqrt{x} d\sqrt{x}) \\
 &= 2 \int u^2 e^{-u} du \\
 &= 2 \left(- \int u^2 d(e^{-u}) \right) && \left| \text{integrate by parts} \right. \\
 &= 2 \left(-u^2 e^{-u} + \int e^{-u} d(u^2) \right) \\
 &= 2 \left(-u^2 e^{-u} + \int 2u e^{-u} du \right) \\
 &= 2 \left(-u^2 e^{-u} - \int 2u d e^{-u} \right) && \left| \text{integrate by parts again} \right. \\
 &= 2 \left(-u^2 e^{-u} - 2u e^{-u} + \int 2e^{-u} du \right) \\
 &= 2 \left(-u^2 e^{-u} - 2u e^{-u} - 2e^{-u} \right) + C \\
 &= 2 \left(-x e^{-\sqrt{x}} - 2\sqrt{x} e^{-\sqrt{x}} - 2e^{-\sqrt{x}} \right) + C
 \end{aligned}$$

Therefore

$$\begin{aligned}
 \int_0^{\infty} \sqrt{x} e^{-\sqrt{x}} dx &= \lim_{t \rightarrow \infty} 2 \left[-x e^{-\sqrt{x}} - 2\sqrt{x} e^{-\sqrt{x}} - 2e^{-\sqrt{x}} \right]_0^{\infty} \\
 &= 4 + \lim_{t \rightarrow \infty} 4 \left(-t e^{-\sqrt{t}} - \sqrt{t} e^{-\sqrt{t}} - e^{-\sqrt{t}} \right) && \left| \text{Set } u = \sqrt{t} \right. \\
 &= 4 - 4 \lim_{u \rightarrow \infty} (u^2 e^{-u} + u e^{-u} + e^{-u}) \\
 &= 4 - 4 \lim_{u \rightarrow \infty} \frac{u^2 + u + 1}{e^u} && \left| \text{use L'Hospital's rule for limit, see below} \right. \\
 &= 4,
 \end{aligned}$$

and the integral converges to 4. In the above computation we used the following limit computation

$$\begin{aligned} \lim_{u \rightarrow \infty} \frac{u^2 + u + 1}{e^u} &= \lim_{u \rightarrow \infty} \frac{2u + 1}{e^u} && \left| \text{Apply L'Hospital's rule} \right. \\ &= \lim_{u \rightarrow \infty} \frac{2}{e^u} \\ &= 0 \end{aligned}$$

Solution. 41.s The integrand is a rational function and therefore we can solve this problem by finding the indefinite integral and then computing the limit. We would need to start by factoring $x^4 + 2$ into irreducible quadratic factors - that is already quite laborious:

$$x^4 + 2 = \left(x^2 + \sqrt[4]{8}x + \sqrt{2}\right) \left(x^2 - \sqrt[4]{8}x + \sqrt{2}\right) \quad .$$

The problem asks us only to establish the convergence of the integral; it does not ask us to compute its actual numerical value.

Therefore we can give a much simpler solution. The function is even and therefore it suffices to establish whether $\int_0^\infty \frac{x^2}{x^4 + 2} dx$ is convergent.

We have that

$$\int_0^\infty \frac{x^2}{x^4 + 2} dx = \int_0^1 \frac{x^2}{x^4 + 2} dx + \int_1^\infty \frac{x^2}{x^4 + 2} dx \quad .$$

The function $\frac{x^2}{x^4 + 2}$ is continuous so $\int_0^1 \frac{x^2}{x^4 + 2} dx$ integrates to a number, which does not affect the convergence of the above expression. Therefore the convergence of our integral is governed by the convergence of $\int_1^\infty \frac{x^2}{x^4 + 2} dx$. To establish that that integral is convergent, we use the comparison theorem as follows.

$$\begin{aligned} \int_1^\infty \frac{x^2}{x^4 + 2} dx &\leq \int_1^\infty \frac{x^2}{x^4} dx && \left| \begin{array}{l} \text{we have that } x^4 + 2 > x^4 \\ \text{and therefore } \frac{x^2}{x^4 + 2} \leq \frac{x^2}{x^4} \end{array} \right. \\ &= \int_1^\infty x^{-2} dx \\ &= \lim_{t \rightarrow \infty} \left[-\frac{1}{x} \right]_1^t \\ &= \lim_{t \rightarrow \infty} 1 - \frac{1}{t} \\ &= 1 \end{aligned}$$

In this way we showed $\int_1^\infty \frac{x^2}{x^4 + 2} dx \leq 1$. Therefore, as $\frac{x^2}{x^4 + 2} \geq 0$ is positive, we can apply the comparison theorem to get that $\int_1^\infty \frac{x^2}{x^4 + 2} dx$ is convergent.

42. Determine whether the integral is convergent or divergent. Motivate your answer. The answer key has not been proofread, use with caution.

(a) $\int_0^\infty \sin x^2 dx$ (This problem is more difficult and may require knowledge of sequences to solve).

ANSWER: CONVERGENT

43. Match the graphs of the parametric equations $x = f(t)$, $y = g(t)$ with the graph of the parametric curve $\gamma : \begin{cases} x = f(t) \\ y = g(t) \end{cases}$

(a)

answer: matches to 43c

(a)

(b)

answer: matches to 43a

(b)

(c)

answer: matches to 43b

(c)

Match the graph of the curve to its graph in polar coordinates and to its polar parametric equations.

(a)	answer: matches 4-b, 4-iv	(a)	(i) $r = 1 + \sin(\theta) + \cos(\theta)$
(b)	answer: matches 4-c, 4-iv	(b)	(ii) $r = \theta, \theta \in [-\pi, \pi]$.
(c)	answer: matches 4-c, 4-iii	(c)	(iii) $r = \cos(3\theta), \theta \in [0, 2\pi]$.
(d)	answer: matches 4-f, 4-iii	(d)	(iv) $r = \frac{1}{4}\sqrt{\theta}, \theta \in [0, 10\pi]$.
(e)	answer: matches 4-d, 4-ii	(e)	(v) $r = 2 + \sin(5\theta)$.
(f)	answer: matches 4-a, 4-iv	(f)	(vi) $r = 2 + \cos(3\theta)$.

45.

- Sketch the curve given in polar coordinates by $r = 2 \sin \theta$. What kind of a figure is this curve? Find an equation satisfied by the curve in the (x, y) -coordinates.
- Sketch the curve given in polar coordinates by $r = 4 \cos \theta$. What kind of a figure is this curve? Find an equation satisfied by the curve in the (x, y) -coordinates.
- Sketch the curve given in polar coordinates by $r = 2 \sec \theta$. What kind of a figure is this curve? Find an equation satisfied by the curve in the (x, y) -coordinates.
- Sketch the curve given in polar coordinates by $r = 2 \csc \theta$. What kind of a figure is this curve? Find an equation satisfied by the curve in the (x, y) -coordinates.
- Sketch the curve given in polar coordinates by $r = 2 \sec(\theta + \frac{\pi}{4})$. What kind of a figure is this curve? Find an equation satisfied by the curve in the (x, y) -coordinates.
- Sketch the curve given in polar coordinates by $r = 2 \csc(\theta + \frac{\pi}{6})$. What kind of a figure is this curve? Find an equation satisfied by the curve in the (x, y) -coordinates.

Solution. 45.c. Recall from trigonometry that if we draw a unit circle as shown below, $\sec \theta$ is given by the signed distance as indicated on the figure. Therefore it is clear that the curve given in polar coordinates by $y = \sec \theta$ is the vertical line passing through $x = 1$. Analogous considerations can be made for a circle of radius 2, from where it follows that $y = 2 \sec \theta$ is the vertical line passing through $x = 2$.

Alternatively, we can find an equation in the (x, y) -coordinates of the curve by the direct computation:

$$x = r \cos \theta = 2 \sec \theta \cos \theta = 2 \quad .$$

Solution. 45.e.

Approach I. Adding an angle α to the angle polar coordinate of a point corresponds to rotating that point counterclockwise at an angle α about the origin. Therefore a point P with polar coordinates $P(2 \sec(\theta + \frac{\pi}{4}), \theta)$ is obtained by rotating at an angle $-\frac{\pi}{4}$ the point Q with polar coordinates $Q(2 \sec(\theta + \frac{\pi}{4}), \theta + \frac{\pi}{4})$. The point P lies on the curve with equation $r = 2 \sec(\theta + \frac{\pi}{4})$ and the point Q lies on the curve with equation $r = 2 \sec \theta$ - the latter curve is the curve from problem 45.c. Thus the curve in the current problem is obtained by rotating the curve from 45.c at an angle of $-\frac{\pi}{4}$. As the curve in Problem 45.c is the vertical line $x = 2$, the curve in the present problem is also a line. Rotation at an angle of $-\frac{\pi}{4}$ of a vertical line yields a line with slope 1. When $\theta = 0$, $x = \frac{2}{\frac{\sqrt{2}}{2}} = 2\sqrt{2}$, $y = 0$ and the curve passes through $(2\sqrt{2}, 0)$. We know the slope of a line and a point through which it passes; therefore the (x, y) -coordinates of our curve satisfy

$$y = x - 2\sqrt{2} \quad .$$

Approach II. We compute

$x = r \cos \theta = \frac{2 \cos \theta}{\cos(\theta + \frac{\pi}{4})}$	multiply by $\cos(\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$
$y = r \sin \theta = \frac{2 \sin \theta}{\cos(\theta + \frac{\pi}{4})}$	multiply by $-\sin(\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$
$x \cos(\frac{\pi}{4}) - y \sin(\frac{\pi}{4}) = 2 \frac{\cos \theta \cos(\frac{\pi}{4}) - \sin \theta \sin(\frac{\pi}{4})}{\cos(\theta + \frac{\pi}{4})}$	add the above
$\frac{\sqrt{2}}{2}(x - y) = 2 \frac{\cos(\theta + \frac{\pi}{4})}{\cos(\theta + \frac{\pi}{4})} = 2$	use $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
$y = x - 2\sqrt{2},$	

and therefore our curve is the line given by the equation above.

46. Find the values of the parameter t for which the curve has horizontal and vertical tangents.

(a) $x = t^2 - t + 1, y = t^2 + t - 1$

(c) $x = \cos(t), y = \sin(3t)$

(b) $x = t^3 - t^2 - t + 1, y = t^2 - t - 1.$

(d) $x = \cos(t) + \sin(t), y = \sin(t).$

47. Show that the parametric curve has multiple tangents at the point and find their slopes.

(a) $x = \cos t, y = 2 \sin(2t)$, two tangents at $(x, y) = (0, 0)$.

of tangents.

(b) $x = \cos t \sin(3t), y = \sin(t) \sin(3t)$, six tangents at $(x, y) = (0, 0)$.

(c) $x = \cos t, y = \sin(3t)$, find the two points at which the curve has double tangent and find the slopes of both pairs

(d) $x = t^3 - t^2 - t + 1, y = t^2 - t - 1$, find a point where the curve has double tangent and find the slopes of the tangents.

48. Find the length of the curve.

(a) $y = x^2, x \in [1, 2]$.

(b) $y = \sqrt{x}, x \in [1, 2]$.

(c) $x = \sqrt{t} - 2t$ and $y = \frac{8}{3}t^{\frac{3}{4}}$ from $t = 1$ to $t = 4$.

(d) $\gamma : \begin{cases} x(t) = \frac{1}{t} + \frac{t^3}{3} \\ y(t) = 2t \end{cases}, t \in [1, 2]$

(e) $\gamma : \begin{cases} x(t) = \frac{1}{t} + t \\ y(t) = 2 \ln t \end{cases}, t \in [1, 2]$

(f) One arch of the cycloid

$$\gamma : \begin{cases} x(t) = t - \sin t \\ y(t) = 1 - \cos t \end{cases}, t \in [0, 2\pi]$$

(g) The cardioid

$$\gamma : \begin{cases} x(t) = (1 + \sin t) \cos t \\ y(t) = (1 + \sin t) \sin t \end{cases}, t \in [0, 2\pi]$$

Solution. 48.a The length of the parametric curve is given by

$$\begin{aligned} L &= \int_1^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\ &= \int_{x=1}^{x=2} \sqrt{1 + 4x^2} dx \\ &= \int_{u=2}^{u=4} \sqrt{u^2 + 1} \left(\frac{1}{2} du\right) \\ &= \frac{1}{2} \int_{u=2}^{u=4} \sqrt{u^2 + 1} du \\ &= \frac{1}{4} [u\sqrt{u^2 + 1} + \ln(u + \sqrt{u^2 + 1})]_2^4 \\ &= \sqrt{17} + \frac{1}{4} \log(\sqrt{17} + 4) - \frac{1}{4} \log(\sqrt{5} + 2) - \frac{\sqrt{5}}{2} \\ &\approx 3.167841 \end{aligned}$$

Substitute $2x = u$
 $dx = \frac{1}{2} du$

$\int \sqrt{u^2 + 1} du$
 $= \frac{1}{2} (u\sqrt{u^2 + 1} + \ln(u + \sqrt{u^2 + 1})) + C$
previously studied

Solution. 48.b

Solution I. The curve can be rewritten in the form $x = y^2$, $y \in [1, \sqrt{2}]$.

$$\begin{aligned}
 L &= \int_1^{\sqrt{2}} \sqrt{\left(\frac{dx}{dy}\right)^2 + 1} \, dy \\
 &= \int_{y=1}^{y=\sqrt{2}} \sqrt{4y^2 + 1} \, dy & \left| \begin{array}{l} \text{Substitute } 2y = u \\ dy = \frac{1}{2} du \end{array} \right. \\
 &= \int_{u=2}^{u=2\sqrt{2}} \sqrt{u^2 + 1} \left(\frac{1}{2} du\right) \\
 &= \frac{1}{2} \int \sqrt{u^2 + 1} \, du & \left| \begin{array}{l} \int \sqrt{u^2 + 1} \, du \\ = \frac{1}{2} \left(u\sqrt{u^2 + 1} + \ln \left(u + \sqrt{u^2 + 1} \right) \right) + C \\ \text{previously studied} \end{array} \right. \\
 &= \frac{1}{4} \left[u\sqrt{u^2 + 1} + \ln \left(u + \sqrt{u^2 + 1} \right) \right]_2^{2\sqrt{2}} \\
 &= \frac{3}{2}\sqrt{2} + \frac{1}{4} \ln \left(2\sqrt{2} + 3 \right) - \frac{1}{4} \ln \left(\sqrt{5} + 2 \right) - \frac{\sqrt{5}}{2} \\
 &\approx 1.083
 \end{aligned}$$

Solution II. The length of the parametric curve is given by

$$\begin{aligned}
 L &= \int_1^2 \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \\
 &= \int_1^2 \sqrt{1 + \left(\frac{1}{2\sqrt{x}}\right)^2} dx \\
 &= \int_{x=1}^{x=2} \sqrt{1 + \frac{1}{4x}} dx & \left| \begin{array}{l} \text{Substitute } 4x = u \\ dx = \frac{1}{4} du \end{array} \right. \\
 &= \int_{u=4}^{u=8} \sqrt{1 + \frac{1}{u}} \left(\frac{1}{4} du\right) \\
 &= \frac{1}{4} \int_4^8 \sqrt{\frac{u+1}{u}} du \\
 &= \frac{1}{4} \int_4^8 \sqrt{\frac{u(u+1)}{u^2}} du \\
 &= \frac{1}{4} \int_4^8 \frac{\sqrt{u^2+u}}{u} du \\
 &= \frac{1}{4} \int_4^8 \frac{\sqrt{u^2+u+\frac{1}{4}-\frac{1}{4}}}{u} du \\
 &= \frac{1}{4} \int_4^8 \frac{\sqrt{\left(u+\frac{1}{2}\right)^2 - \frac{1}{4}}}{u} du \\
 &= \frac{1}{4} \int_4^8 \frac{\sqrt{\frac{1}{4}((2u+1)^2 - 1)}}{u} du & \left| \begin{array}{l} \text{Substitute } 2u+1 = z \\ u = \frac{z-1}{2} \\ du = \frac{1}{2} dz \end{array} \right. \\
 &= \frac{1}{8} \int_{u=4}^{u=8} \frac{\sqrt{(2u+1)^2 - 1}}{u} du \\
 &= \frac{1}{8} \int_{z=9}^{z=17} \frac{\sqrt{z^2-1}}{\frac{z-1}{2}} \frac{1}{2} dz \\
 &= \frac{1}{8} \int_{z=9}^{z=17} \frac{\sqrt{z^2-1}}{z-1} dz & \left| \begin{array}{l} \text{Trig. subst.: } z = \sec \theta \\ \sqrt{z^2-1} = \tan \theta \\ dz = \sec \theta \tan \theta d\theta \end{array} \right. \\
 &= \frac{1}{8} \int_{\theta=\text{arcsec}(9)}^{\theta=\text{arcsec}(17)} \frac{\tan \theta}{\sec \theta - 1} \sec \theta \tan \theta d\theta \\
 &= \frac{1}{8} \int_{\alpha}^{\beta} \frac{\tan^2 \theta}{\sec \theta - 1} \sec \theta d\theta \\
 &= \frac{1}{8} \int_{\alpha}^{\beta} \frac{\sec^2 \theta - 1}{\sec \theta - 1} \sec \theta d\theta \\
 &= \frac{1}{8} \int_{\alpha}^{\beta} \frac{(\sec \theta - 1)(\sec \theta + 1)}{\sec \theta - 1} \sec \theta d\theta \\
 &= \frac{1}{8} \int_{\alpha}^{\beta} (\sec^2 \theta + \sec \theta) d\theta & \left| \begin{array}{l} \text{Use } \tan^2 \theta = \sec^2 \theta - 1 \\ \int \sec \theta d\theta = \ln |\sec \theta + \tan \theta| + C \\ \text{previously studied} \end{array} \right. \\
 &= \frac{1}{8} [\tan \theta + \ln |\sec \theta + \tan \theta|]_{\alpha}^{\beta} \\
 &= \frac{1}{8} (12\sqrt{2} + \ln(17 + 12\sqrt{2}) - 4\sqrt{5} - \ln(9 + 4\sqrt{5})) \\
 &= \frac{1}{8} \ln(12\sqrt{2} + 17) - \frac{1}{8} \ln(4\sqrt{5} + 9) - \frac{\sqrt{5}}{2} + \frac{3}{2}\sqrt{2} \\
 &\approx 1.083 .
 \end{aligned}$$

The two answers are both approximately 1.083, so that serves to cross verify our two solutions against one another.

Comparing the two answers we notice that the logarithmic parts in the two answers look different (yet they must be equal). It follows that

$$\frac{1}{8} \ln(12\sqrt{2} + 17) - \frac{1}{8} \ln(4\sqrt{5} + 9) = \frac{1}{4} \ln(2\sqrt{2} + 3) - \frac{1}{4} \ln(\sqrt{5} + 2) .$$

A short computation (which computation?), left to the reader, confirms that indeed those two expressions are equal.

Solution. 48.c. The length of the parametric curve is given by

$$L = \int_1^4 \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt .$$

We have that

$$\begin{aligned} \frac{dx}{dt} &= \frac{1}{2\sqrt{t}} - 2 \\ \frac{dy}{dt} &= 2t^{-\frac{1}{4}} \\ \left(\frac{dx}{dt}\right)^2 &= \frac{1}{4t} - \frac{2}{\sqrt{t}} + 4 \\ \left(\frac{dy}{dt}\right)^2 &= 4t^{-\frac{1}{2}} = \frac{4}{\sqrt{t}} \\ \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 &= \frac{1}{4t} + 2\frac{1}{\sqrt{t}} + 4 = \left(\frac{1}{2\sqrt{t}} + 2\right)^2 . \end{aligned}$$

$\frac{1}{2\sqrt{t}} + 2$ is positive and $\sqrt{\left(\frac{1}{2\sqrt{t}} + 2\right)^2} = \frac{1}{2\sqrt{t}} + 2$. So the integral becomes

$$L = \int_1^4 \left(\frac{1}{2\sqrt{t}} + 2\right) dt = \left[\sqrt{t} + 2t\right]_{t=1}^{t=4} = (2 + 8) - (1 + 2) = 7 .$$

49. Set up an integral that expresses the length of the curve and find the length of the curve.

(a) $\begin{cases} x(t) = e^t + e^{-t} \\ y(t) = 5 - 2t \end{cases}, t \in [0, 3]$

ANSWER: $e^3 - e - 8$

(b) $\begin{cases} x(t) = \sin t + \cos t \\ y(t) = \sin t - \cos t \end{cases}, t \in [0, \pi]$

ANSWER: $2\sqrt{2}\pi$

50. Give a geometric definition of the cycloid curve using a circle of radius 1. Using that definition, derive equations for the cycloid curve. Find area locked between one “arch” of the cycloid curve and the x axis.

51. (a) The curve given in polar coordinates by $r = 1 + \sin 2\theta$ is plotted below by computer. Find the area lying outside of this curve and inside of the circle $x^2 + y^2 = 1$.

ANSWER: $2 - \frac{\pi}{2}$

(b) The curve given in polar coordinates by $r = \cos(2\theta)$ is plotted below by computer. Find the area lying inside the curve and outside of the circle $x^2 + y^2 = \frac{1}{4}$.

ANSWER: $\frac{9}{16} + \frac{\pi}{8}$

(c) Below is a computer generated plot of the curve $r = \sin(2\theta)$. Find the area locked inside one petal of the curve and outside of the circle $x^2 + y^2 = \frac{1}{4}$.

ANSWER: $\frac{\pi}{8} + \frac{91}{64}$

Solution. 51.a. A computer generated plot of the two curves is included below. The circle $x^2 + y^2 = 1$ has one-to-one polar representation given by $r = 1, \theta \in [0, 2\pi)$. Except the origin, which is traversed four times by the curve $r = 1 + \sin(2\theta)$, the second curve is in a one-to-one correspondence with points in the r, θ -plane given by the equation $r = 1 + \sin(2\theta), \theta \in [0, 2\pi)$. Since the two curves do not meet in the origin, we may conclude that the two curves may intersect only when their values for r and θ coincide. Therefore we have an intersection when

$$\begin{aligned} 1 + \sin(2\theta) &= 1 \\ \sin(2\theta) &= 0 \\ \theta &= 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2} \quad | \text{ because } \theta \in [0, 2\pi) \end{aligned}$$

Therefore the two curves meet in the points $(0, 1)(-1, 0)$ and $(0, -1), (1, 0)$.

Denote the investigated region by A . From the computer-generated plot, it is clear that when a point has polar coordinates $\theta \in [\frac{\pi}{2}, \pi] \cup [\frac{3\pi}{2}, 2\pi]$, $r \in [1 + \sin(2\theta), 1]$ it lies in A . Furthermore, the points r, θ lying in the above intervals are in one-to-one correspondence with the points in A .

Suppose we have a curve $r = f(\theta)$, $\theta \in [a, b]$ for which no two points lie on the same ray from the origin. Recall from theory that the area swept by that curve is given by

$$\int_a^b \frac{1}{2} f^2(\theta) d\theta \quad .$$

Therefore the area a of A is computed via the integrals

$$\begin{aligned} a &= \int_{\frac{\pi}{2}}^{\pi} \frac{1}{2} \left(\underbrace{1}_{\text{outer curve}}^2 - \left(\underbrace{1 + \sin(2\theta)}_{\text{inner curve}} \right)^2 \right) d\theta + \int_{\frac{3\pi}{2}}^{2\pi} \frac{1}{2} (1^2 - (1 + \sin(2\theta))^2) d\theta & \left| \begin{array}{l} \text{use the symmetry of } A \\ \\ \text{use } \sin^2 z = \frac{1 - \cos(2z)}{2} \end{array} \right. \\ &= \int_{\frac{\pi}{2}}^{\pi} (1^2 - (1 + \sin(2\theta))^2) d\theta = \int_{\frac{\pi}{2}}^{\pi} (-2\sin(2\theta) - \sin^2(2\theta)) d\theta \\ &= \int_{\frac{\pi}{2}}^{\pi} \left(-2\sin(2\theta) - \frac{1}{2} + \frac{1}{2} \cos(4\theta) \right) d\theta = \left[\cos(2\theta) - \frac{1}{2}\theta - \frac{1}{8} \sin(4\theta) \right]_{\frac{\pi}{2}}^{\pi} \\ &= 2 - \frac{\pi}{4} \quad . \end{aligned}$$

Solution. 51.b A computer generated plot of the figure is included below. The circle $x^2 + y^2 = \frac{1}{4}$ is centered at 0 and of radius $\frac{1}{2}$ and therefore can be parametrized in polar coordinates via $r = \frac{1}{2}$, $\theta \in [0, 2\pi]$.

Points with polar coordinates (r_1, θ_1) and (r_2, θ_2) coincide if one of the three holds:

- $r_1 = r_2 \neq 0$ and $\theta_1 = \theta_2 + 2k\pi$, $k \in \mathbb{Z}$,
- $r_1 = -r_2 \neq 0$ and $\theta_1 = \theta_2 + (2k + 1)\pi$, $k \in \mathbb{Z}$,
- $r_1 = r_2 = 0$ and θ is arbitrary.

To find the intersection points of the two curves we have to explore each of the cases above. The third case is not possible as the circle does not pass through the origin. Suppose we are in the first case. Then the value of r (as a function of θ) is equal for the two curves. Thus the two curves intersect if

$$\begin{aligned} r = \cos(2\theta) &= \frac{1}{2} \\ 2\theta &= \pm \frac{\pi}{3} + 2k\pi & \left| \begin{array}{l} \text{where } k \in \mathbb{Z} \\ \text{where } k \in \mathbb{Z} \\ \text{all other values discarded as } \theta \in [0, 2\pi] \end{array} \right. \\ \theta &= \pm \frac{\pi}{6} + k\pi \\ \theta &= \frac{\pi}{6}, \frac{\pi}{6} + \pi, \frac{-\pi}{6} + \pi, \frac{-\pi}{6} + 2\pi \\ \theta &= \frac{\pi}{6}, \frac{7\pi}{6}, \frac{5\pi}{6}, \frac{11\pi}{6} \end{aligned}$$

This gives us only four intersection points, and the computer-generated plot shows eight. Therefore the second case must yield new intersection points: the two curves intersect also when

$$\begin{aligned} r = \cos(2\theta) &= -\frac{1}{2} \\ 2\theta &= \pm \frac{2\pi}{3} + 2k\pi & \left| \begin{array}{l} \text{where } k \in \mathbb{Z} \\ \text{where } k \in \mathbb{Z} \\ \text{all other values are discarded as } \theta \in [0, 2\pi] \end{array} \right. \\ \theta &= \pm \frac{\pi}{3} + k\pi \\ \theta &= \frac{\pi}{3}, \frac{\pi}{3} + \pi, \frac{-\pi}{3} + \pi, \frac{-\pi}{3} + 2\pi \\ \theta &= \frac{\pi}{3}, \frac{4\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{3} \quad . \end{aligned}$$

From the computer-generated plot below, we can see that the area we are looking for is 4 times the area locked between the two curves for $\theta \in [-\frac{\pi}{6}, \frac{\pi}{6}]$. Therefore the area we are looking for is given by

$$4 \int_{-\frac{\pi}{6}}^{\frac{\pi}{6}} \frac{1}{2} \left(\cos^2(2\theta) - \left(\frac{1}{2} \right)^2 \right) d\theta \quad .$$

We leave the above integral to the reader.

Solution. 51.c. The circle $x^2 + y^2 = \frac{1}{4}$ is centered at 0 and of radius $\frac{1}{2}$ and therefore can be parametrized in polar coordinates via $r = \frac{1}{2}, \theta \in [0, 2\pi)$.

Points with polar coordinates (r_1, θ_1) and (r_2, θ_2) coincide if one of the three holds:

- $r_1 = r_2 \neq 0$ and $\theta_1 = \theta_2 + 2k\pi, k \in \mathbb{Z}$,
- $r_1 = -r_2 \neq 0$ and $\theta_1 = \theta_2 + (2k+1)\pi, k \in \mathbb{Z}$,
- $r_1 = r_2 = 0$ and θ is arbitrary.

To find the intersection points of the two curves we have to explore each of the cases above. The third case is not possible as the circle does not pass through the origin. Suppose we are in the first case. Then the value of r (as a function of θ) is equal for the two curves. Thus the two curves intersect if

$$\begin{array}{rcl} r = \sin(2\theta) & = & \frac{1}{2} \\ 2\theta & = & \frac{\pi}{6} + 2k\pi \text{ or } \frac{5\pi}{6} + 2k\pi \\ \theta & = & \frac{\pi}{12} + k\pi \text{ or } \frac{5\pi}{12} + k\pi \\ \theta & = & \frac{\pi}{12}, \frac{13\pi}{12}, \frac{5\pi}{12}, \frac{17\pi}{12} \end{array} \quad \left| \begin{array}{l} \text{where } k \in \mathbb{Z} \\ \text{where } k \in \mathbb{Z} \\ \text{other values discarded as} \\ \theta \in [0, 2\pi] \end{array} \right.$$

This gives us only four intersection points, and the computer-generated plot shows eight. Therefore the second case must yield 4 new intersection points. However, from the figure we see there are only two intersection points that participate in the boundary of our area, and both of those were found above. Therefore we shall not find the remaining 4 intersections.

Both the areas locked by the petal and the area locked by the section of the circle are found by the formula for the area locked by a polar curve. Subtracting the two we get that the area we are looking for is:

$$\begin{aligned} \text{Area} &= \int_{\theta=\frac{\pi}{12}}^{\theta=\frac{5\pi}{12}} \frac{1}{2} \left(\sin^2(2\theta) - \left(\frac{1}{2}\right)^2 \right) d\theta \\ &= \frac{1}{2} \int_{\theta=\frac{\pi}{12}}^{\theta=\frac{5\pi}{12}} \left(\frac{1 - \cos(4\theta)}{2} - \frac{1}{4} \right) d\theta \\ &= \frac{1}{2} \left[\frac{1}{4}\theta - \frac{\sin(4\theta)}{8} \right]_{\theta=\frac{\pi}{12}}^{\theta=\frac{5\pi}{12}} \\ &= \frac{\pi}{24} + \frac{\sqrt{3}}{16} . \end{aligned}$$

52. The answer key has not been proofread, use with caution.

- (a) Sketch the graph of the curve given in polar coordinates by $r = 3 \sin(2\theta)$ and find the area of one petal.

answer: $\frac{8}{9}\pi$, curve sketch:

- (b) Sketch the graph of the curve given in polar coordinates by $r = 4 + 3 \sin \theta$ and find the area enclosed by the curve.

answer: $\frac{41}{2}\pi$, curve sketch: